

THE SCATTERING STRUCTURE OF OPEN FJRW THEORY OF $x^r + y^s$

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ABSTRACT. Gross, Kelly and Tessler constructed a genus-zero open enumerative theory for the Landau–Ginzburg model $(x^r + y^s, \mu_r \times \mu_s)$: the open FJRW invariants depend on boundary conditions, and the systems of invariants form a torsor under a wall-crossing group which they observe is “very similar in spirit” to the tropical vertex group of Kontsevich–Soibelman. We turn the observation into a structure theory. Every primary transition between invariant systems has a unique slope-ordered factorization by the boundary rays, to every order in the deformation parameters; for descendents the compatible factors are defined coefficientwise on finite divisor-closed coefficient sets. They define a relative scattering structure on an integral-affine sector of directions. Its pro-nilpotent walls are supported precisely on the deformation monomials that carry a critical graph of positive Euler weight, while marginal critical graphs supply a torus factor, sharpening the central-charge threshold for wall-crossing; two unique extremal systems of invariants produce a canonical relative scattering diagram, and the open topological recursion supplies the direction transverse to its walls, so that the closed theory determines all invariants up to the torsor action. Once descendents are admitted the walls accumulate densely, so open chambers are replaced by coefficientwise slope cuts. The potential family is then the coefficientwise broken-line transport of a unique cut-independent seed system, for a canonical assignment of each potential coefficient to a slope cut. The seeds admit a finite ordered-chain inversion formula and are virtual Euler degrees of compact bar-cascade objects assembled from relative-Euler and critical-homotopy zero loci: the seed values vanish at the extreme coefficients that receive no bending contributions from other initial monomials, a consequence delimited by the interlacing of walls and coefficients and by a monotonicity property of bending (the purely tropical part of the theory, where the invariants are genuine broken-line counts), while the nonzero seed values are non-tropical input, beyond what the diagram determines. Every finite divisor-closed primary or descendent coefficient window is realized, in increasing slope, by a concatenated family of simple canonical homotopies of boundary conditions. Relative versions of the underlying extension arguments make these realizations compatible coefficientwise; staging is forced by a provable obstruction. The two extreme indices admit symmetric polar representatives: on every wall-bearing diagonal, all balanced sections except the surviving endpoint section are nowhere zero. The resulting homotopy-coherent transport on the punctured boundary-charge sector is projective: it factors through the interval of slopes, so radial paths act trivially. Since the punctured sector is contractible, its total extreme-to-extreme product is boundary-to-boundary transport rather than monodromy about the common ray origin. This gives a precise obstruction to recovering a genuine singular affine position space from the present continuation data without an additional valuation or affine gluing. For $x^5 + y^5$ we compute the first wall functions and several new invariants, and verify the recursion in full against exactly computed closed data. Three corrections to the literature accompany the results. All computations are exact, and code reproducing them accompanies the paper.

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1. INTRODUCTION

1.1. The theory. Gross, Kelly and Tessler [GKT22] defined genus-zero open FJRW invariants

$$\nu_{\Gamma, \mathbf{d}} = \left\langle \prod_i \tau_{d_i}^{(a_i, b_i)} \sigma_1^{k_1} \sigma_2^{k_2} \sigma_{12} \right\rangle^{\mathfrak{s}, o}$$

for the Landau–Ginzburg model $(W = x^r + y^s, \mu_r \times \mu_s)$: relative Euler numbers of descendent Witten bundles over moduli of W -spin disks, which are real orbifolds with corners. The invariants depend on a family $\mathfrak{s}$ of canonical multisections (boundary conditions). Three of their results organize this dependence completely. First, the systems of invariants arising from canonical families form a set $\Omega_{r,s}(I)$ (I a finite set of internal marking labels as in [GKT22]; in universal finite-order statements below it is chosen *saturated* at the order under consideration, meaning it carries at least $k - 1$ labels of every primary twist at order k), characterized by explicit polynomial conditions (the *chamber-index* conditions of [GKT22, Def. 3.46, Cor. 5.13]; a *chamber index* is their name for such a system of invariants, not a chamber of any diagram in this paper): normalizations for at most one internal marking, and the conditions $\mathcal{A}(J, \mathbf{d}, \nu) = (-1)^d$ (singletons) and $\mathcal{A}(J, \mathbf{d}, \nu) = 0$ for $d(J, \mathbf{d}) < 0$ (with $d(J, \mathbf{d})$ as in [GKT22, Def. 4.8]), where $\mathcal{A}$ is a Γ -function-weighted polynomial in the invariants. Second, $\Omega_{r,s}(I)$ is a torsor: a wall-crossing group acts on it faithfully and transitively [GKT22, Thm. 4.26], with one generating summand per *critical graph*, the disk type responsible for jumps. Third, the generating-function potential W^ν of any system has oscillatory integrals over the good basis equal to the closed extended FJRW generating function [GKT22, Thm. 0.3], an open mirror theorem building on the closed Landau–Ginzburg mirror symmetry of [LLSS17, HLSW22].

Gross–Kelly–Tessler remark [GKT22, Rem. 0.8] that their wall-crossing group is close in spirit to the tropical vertex group of Kontsevich–Soibelman [KS06], as used for Fano mirror symmetry in [GPS10, Gro10] and in the log Calabi–Yau setting [GHK15], and their survey raises the problem of a tropical formulation of the open theory [GKT26]. Such a formulation is not immediate. A Landau–Ginzburg model carries no target space and hence no SYZ fibration, and on these grounds Maher [Mah24, Rem. 5.1.1] records the combinatorial machinery of the Gross–Siebert program as out of reach for these models. The present paper develops the scattering structure of the theory and computes its first walls. The construction is algebraic, carried out over the wall-crossing group rather than over a degenerating target, which is what makes it available here. Every finite primary or descendent coefficient window is realized by canonical homotopies in Theorem 1.6, compatibly in the coefficientwise limit. The sector itself is the effective cone of the two open-FJRW boundary charges. Theorem 1.7 proves that the induced continuation theory on its punctured cone factors through slope and has no origin loop. Thus the cone is an intrinsic relative scattering sector, but a singular affine position space of the kind supplied by an SYZ fibration requires genuinely additional input.

1.2. Results. Throughout, $\mathfrak{g}^{r,s}$ denotes the Landau–Ginzburg wall-crossing Lie algebra of [GKT22, Def. 4.22]: it is spanned by elements $u_{J, \mathbf{d}} X_{k_1, k_2}$ with

$$X_{k_1, k_2} = x^{k_1} y^{k_2} ((k_2 + 1)x \partial_x - (k_1 + 1)y \partial_y)$$

the Hamiltonian vector field of $x^{k_1+1} y^{k_2+1}$ for $dx \wedge dy$, and $u_{J, \mathbf{d}} = \prod_j t_{a_j, b_j, d_j}$ a monomial in the deformation parameters ($t_{a,b,d}$ dual to the insertion $\tau_d^{(a,b)}$; the descendent index is suppressed when $d = 0$), subject to the congruences $k_1 \equiv r(J) \pmod{r}$, $k_2 \equiv s(J) \pmod{s}$ and the dimension constraint

$$(1) \quad sk_1 + rk_2 = m(J, \mathbf{d}) - rs, \quad m(J, \mathbf{d}) = rs + \sum_j (sa_j + rb_j + rs(d_j - 1)).$$

The right-hand side of (1) is the *Euler weight* $w(J, \mathbf{d})$ of the monomial $u_{J, \mathbf{d}}$. By the classification [GKT22, Not. 2.38], each $(J, \mathbf{d})$ carries an integer count $N = N(J, \mathbf{d})$ (Definition 2.1): when $N \geq 0$ there are $N+1$ balanced disk types (potential coefficients), and when $N \geq 1$ there are N critical graphs (walls); below the respective thresholds those sets are empty. The Lie bidegree of a critical graph is its boundary degree shifted by $(-1, -1)$: the wall Hamiltonian $x^{k_1+1}y^{k_2+1}$ is the boundary monomial $\sigma_1^\# \sigma_2^\#$ of the critical disk, and the shift is the removal of the single fully twisted point σ_{12} that balanced disks carry and critical disks do not. This is the primitive form $dx \wedge dy$ acting as an integral-affine shift (§2).

Theorem 1.1 (Weight trichotomy and the threshold; §2). *In the enlarged algebra of Remark 2.3 (the wall-crossing algebra with the marginal torus directions adjoined), for a nonempty marking multiset J , wall summands over the monomial $u_{J, \mathbf{d}}$ exist if and only if $N(J, \mathbf{d}) \geq 1$ and $w(J, \mathbf{d}) \geq 0$. The pro-nilpotent summands are exactly those of positive weight. For a pro-nilpotent wall, positive weight is necessary but not sufficient (at $r = s = 5$ the monomial $t_{3,3}$ has $w = 5$ but $N = 0$, hence no wall); at $w = 0$ the constraint (1) admits only the marginal torus bidegree $(0, 0)$, and at $w < 0$ it admits nothing. Consequently a monomial of nonpositive weight carries no pro-nilpotent wall generator of its own (although higher-order transport from divisor monomials may reach its coefficients), and the central charge $\hat{c}_W = 2 - \frac{2}{r} - \frac{2}{s}$ governs a trichotomy: pro-nilpotent $d = 0$ wall-crossing exists if and only if $\hat{c}_W > 1$ (we call the pair (r, s) hyperbolic); at $\hat{c}_W = 1$ (parabolic) primary wall-crossing exists but is carried entirely by the marginal torus direction; for $\hat{c}_W < 1$ (simple) there is no primary wall-crossing at all.*

Theorem 1.2 (The relative scattering presentation; §3). *(i) (Loop consistency.) For any family of simple canonical homotopies supplied by [GKT22, Lem. 3.39] with equal endpoints, the time-ordered product of the critical-graph jump automorphisms constructed in the proof of [GKT22, Thm. 5.8] is the identity; for such homotopies with arbitrary endpoints, the total product depends only on the endpoints. (ii) (Boundary-ray presentation.) In the primary algebra, modulo every power of the deformation ideal, every element of the enlarged wall-crossing group factors uniquely as a slope-ordered product of exponentials supported on the shifted boundary rays $\mathbb{R}_{\geq 0}(k_1+1, k_2+1)$; the marginal torus terms belong to the $(1, 1)$ -ray. The compatible factorizations define its completed boundary-ray presentation. In the full descendent algebra the same presentation is taken on finite divisor-closed coefficient sets and passed to the coefficientwise limit. We call the resulting ray-decorated, boundary-to-boundary transport a relative scattering structure: unlike a standard consistent diagram on a complete affine base, its total transport need not be the identity.*

The statements below use the following language. The *diagonal* of a deformation monomial is its set of balanced coefficients, the lattice points of one line in the boundary-degree plane (Definition 2.1). The rest is developed in §7: the *direction sector* is the closed first quadrant of boundary-monomial directions; a *slope cut* is a direction in the sector together with a choice of side when a wall sits on that direction; the transport $\Psi(q)$ to a cut q is the slope-ordered product of the wall factors from the σ_1 -axis side (the *base cut*) up to q ; and a bending contribution to a coefficient is *foreign* if it originates from another monomial's seed (Definitions 7.1 and 7.4).

Theorem 1.3 (Normal forms and the canonical diagram; §5). *On every wall-bearing diagonal, including a marginal diagonal, the wall action on the $N+1$ potential coefficients is triangular; consequently there exist unique chamber indices $\nu^{\min}$ and $\nu^{\max}$ in which, on every such diagonal, only the extreme coefficient survives. The boundary-slope-ordered factorization of the unique group element carrying $W^{\nu^{\min}}$ to $W^{\nu^{\max}}$ is a canonical, choice-free completed primary relative diagram $\mathfrak{D}_0^{r,s}$, computable order by order; its descendent extension $\mathfrak{D}^{r,s}$ is interpreted*

coefficientwise on slope cuts. We write θ_ρ for the factor of the diagram supported on the ray ρ . For $x^5 + y^5$ its walls begin

$$\begin{aligned}\theta_{(1,2)} &= \exp\left(\frac{1}{25} t_{2,3} t_{3,3} X_{0,1} + \cdots\right), & \theta_{(2,1)} &= \exp\left(\frac{1}{25} t_{3,2} t_{3,3} X_{1,0} + \cdots\right), \\ \theta_{(1,1)} &= \exp\left(\frac{1}{50} t_{3,3}^2 X_{1,1} + \cdots\right).\end{aligned}$$

The subscripts label primitive boundary directions. A wall factor collects all critical-graph terms for which (k_1+1, k_2+1) lies on that ray (Proposition 2.2).

Theorem 1.4 (Effective determination; §4). *On each $(J, \mathbf{d})$ -diagonal the wall directions span, in the NS-visible case (both residues in the Neveu–Schwarz range: $r(J) \leq r-2$ and $s(J) \leq s-2$), exactly the kernel of the good-basis period functional, and the chamber-index conditions of [GKT22] amount to one Γ /period-weighted linear condition per diagonal with right-hand side determined by strictly lower order. Together with the characterization $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ of [GKT22, Cor. 5.13] ($\Upsilon_{r,s}(I)$ the set of solutions of the chamber-index conditions) this determines all open FJRW invariants of $x^r + y^s$, up to the torsor action, from the normalizations and the closed extended theory: the topological recursion of [GKT22, Thm. 0.5] appears as the direction transverse to the walls.*

Theorem 1.4 is the algebraic form of the statement that *the scattering structure computes the open theory*. The worked case $x^5 + y^5$ (§5, §6) produces, beyond the wall functions above, the invariant relations and values

$$\begin{aligned}(2) \quad & \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1 \sigma_2^6 \sigma_{12} \rangle + \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1^6 \sigma_2 \sigma_{12} \rangle = -\frac{2}{5}, \\ (3) \quad & \langle \tau_0^{(3,3)} \tau_0^{(3,3)} \tau_0^{(3,3)} \sigma_1^4 \sigma_2^4 \sigma_{12} \rangle = \frac{1}{5}, \quad \langle \tau_0^{(1,2)} \tau_0^{(2,1)} \tau_0^{(2,2)} \sigma_{12} \rangle = -\frac{1}{25},\end{aligned}$$

the first a wall relation (only the sum is invariant; the difference jumps across the first symmetric pro-nilpotent primary wall $t_{3,3}^2 X_{1,1}$), the second and third absolute: they live on diagonals with no walls, the second on a diagonal invisible to the good-basis periods. The seed relation (for the seed system of Theorem 1.5 below) takes the uniform form

$$(4) \quad \frac{\nu_{(r,0)}}{r} + \frac{\nu_{(0,s)}}{s} = -\frac{1}{rs} \quad (\text{e.g. } \langle \sigma_1^r \sigma_{12} \cdots \rangle + \langle \sigma_2^s \sigma_{12} \cdots \rangle = -\frac{1}{r} \text{ at } r=s),$$

for every two-insertion seed diagonal $\{(a_1, b_1), (a_2, b_2)\}$, $|J| = 2$, with $a_1 + a_2 = r$, $b_1 + b_2 = s$ (Remark 4.3). Here $\nu_{(k_1, k_2)}$ denotes the invariant with boundary profile $\sigma_1^{k_1} \sigma_2^{k_2} \sigma_{12}$, and the suppressed insertions are the two interior markings of the seed.

The last three results are geometric: the transport representation, the finite-support and coefficientwise realization of the diagram by homotopies of boundary conditions, and the exact position-space obstruction carried by that realization.

Theorem 1.5 (The transport representation; §7). *Fix the anchor assignment: each monomial $u_{J, \mathbf{d}} x^{k_1} y^{k_2}$ of the potential is anchored at the one-sided slope cut through its direction (k_1, k_2) , on the side toward the $(1, 1)$ -direction. A constant monomial is anchored at the base cut; on slope 1 either one-sided convention gives the same coefficient. Then, by induction in the divisibility order of deformation monomials: on every monomial $u_{J, \mathbf{d}}$ the difference between the potential's coefficient vector and the transported contributions of the seeds of the proper divisor monomials is independent of the cut, and defining the seed system by these constants, every cut potential is the coefficientwise transport of the seeds,*

$$W(q) = \sum_m \Psi(q) \Psi(q_m)^{-1} (\text{seed}_m),$$

the sum over the monomials m of the potential and q_m the anchor cut of m . On the coefficient of a fixed deformation monomial u , project every factor to the finite divisor-closed quotient consisting of the divisors of u ; only finitely many projected boundary-ray factors are nontrivial, and their slope-ordered product defines $\Psi(q)$ coefficientwise from the base cut to q . The seed system is the unique one with this property. On a coefficient receiving no foreign bending, the seed equals its anchored value; for a foreign-free extreme coefficient of a wall-bearing diagonal that value is its vanishing extreme-index value (Proposition 7.3), so every such coefficient carries seed zero. Those zero-seed coefficients form the purely tropical part of the theory, while the nonzero seed values are non-tropical input.

Theorem 1.5 is the structural core of the tropical description of the open theory: the invariants are computed by broken lines (monotone transports bending at the walls of $\mathfrak{D}^{r,s}$) from canonical seed data. The seeds play the role of the initial monomials of broken lines in [Gro10, CPS24]; the word carries no cluster-theoretic meaning here. The ordered-chain formula and corrected relative-Euler cascade cycles below give them a zero-dimensional enumerative interpretation. The two extreme indices have symmetric polar representatives whose nonextreme balanced sections are nowhere zero (Theorem 7.29), and the cascade cycles are the virtual Euler cycles of single compact Kuranishi bar realizations (Theorem 7.14). Theorem 1.7 determines the exact boundary of the position-space reading: the geometric continuation exists on the punctured sector but factors through slope and cannot supply origin monodromy. The whole is the Landau–Ginzburg analogue of the tropical disk-counting picture of [Gro10, CPS24].

Theorem 1.6 (Finite-support and pro-geometric realization; §7.3). *Let S be a finite divisor-closed set of deformation monomials containing 1. Choose a finite labeled set I with at least*

$$n_{a,b}(S) := \max_{u \in S} \sum_{d \geq 0} \text{ord}_{t_{a,b,d}}(u)$$

labels of every twist (a, b) occurring in S . The quotient diagram has finitely many rays $\rho_1, \dots, \rho_n$, listed in increasing slope. There is a canonical homotopy of boundary conditions, obtained by concatenating families of simple canonical homotopies in the sense of [GKT22, §3.4], from a family realizing $\nu_I^{\min}$ to one realizing $\nu_I^{\max}$, the restrictions of the universal extreme indices, such that after reduction to the quotient supported on S every nonzero jump is supported on a single boundary ray, the rays carrying nonzero reduced jumps occur in increasing slope as the homotopy time increases, and the total reduced jump on ρ_i is the wall automorphism θ_{ρ_i} of Theorem 1.3. These realizations can be chosen compatibly over a cofinal construction-closed exhaustion of the labeled graph/descendent data. The resulting coefficientwise homotopy has pro-symmetric endpoints $\nu^{\min}, \nu^{\max}$, and every fixed graph and descendent vector sees only finitely many nonconstant ray stages.

The staging is forced: above the central-charge threshold a single normal-form family of [GKT22, Lem. 3.39] provably cannot separate the walls of a diagonal carrying two of them, since they all jump at one time (Proposition 7.23), and the exact first order at which two walls share a diagonal is computed for every hyperbolic pair (Proposition 7.21, Remark 7.26). Realizing each boundary-ray factor in its own stage is what evades the obstruction.

Theorem 1.7 (Projective coherence and the position-space obstruction; §7.4). *For every finite divisor-closed coefficient set S , the staged realization extends to a homotopy-coherent transport pseudofunctor on the punctured sector $C_{\partial}^{\times} = C_{\partial} \setminus \{0\}$: points give canonical multisection families, tame paths give concatenated canonical homotopies, and tame disks give relative two-parameter families. An oriented crossing of a wall ρ induces $\theta_{\rho,S}^{\pm 1}$. This pseudofunctor*

factors, up to coherent equivalence, through

$$\pi : C_{\partial}^{\times} \longrightarrow \mathbb{P}(C_{\partial}),$$

and the factorizations are compatible coefficientwise as S grows. Consequently every radial path acts trivially and dilation changes no GKT transport observable. Moreover $C_{\partial}^{\times} \cong (0, \infty) \times [0, 1]$ is contractible: the product $\prod_{\rho}^{\rightarrow} \theta_{\rho} = g^{\min \rightarrow \max}$ is transport between the two boundary sides, not origin monodromy. Hence the present GKT continuation data do not intrinsically determine the radially detected modulus, singular affine extension, or vertex monodromy required of the proposed position space. An affirmative Gross–Siebert/SYZ and theta-function theory requires additional valuation, affine-gluing, and binary enumerative input.

1.3. Three corrections to the literature. Our computations bear on three points in [GKT22, GKT26], recorded here and in Remarks 4.3 and 2.3 with the supporting derivations. First, imposing the open mirror theorem [GKT22, Thm. 0.3] directly on the oscillatory expansion of the potential — the mechanism of their Theorem 0.2 — gives the seed values (4); at $r = s = 4$ this yields $-\frac{1}{4}$ for the sum in [GKT26, Ex. 5.5], where the value $-\frac{1}{8}$ is stated. The partition count in the expansion of $\mathcal{A}$ forced by [GKT22, Thm. 4.9] (ordered partitions of the labeled marking set, with $1/h!$) also reproduces, through the proven [GKT22, Thm. 5.4], the closed value $\mathcal{A}(\{(1, 1), (1, 1)\}, 0) = \langle \tau_0^{(1,1)3} \rangle = 1$ at $r = s = 5$, pinning the same count independently. Second, at wall-bearing marginal monomials the critical graphs have torus bidegree $(0, 0)$, which the pro-nilpotent algebra of [GKT22, Def. 4.22] excludes; the proofs of [GKT22, Thm. 5.8, Cor. 5.13] nevertheless use exactly these summands (through their Q_0 -sets and Lemma 5.16), so the group there should be read as enlarged by the marginal torus summands, the elliptic wall-crossing group appearing in Maher’s thesis [Mah24, §5.2.2]. All statements in this paper use the enlarged group. Third, in the same proof, equation (5.51) of [GKT22] carries the coefficient $s k_2(\Lambda_{J,1})$ where the $p = 0$ instance of its own (5.52) gives $s k_1(\Lambda_{J,1})$; the latter is the one consistent with the action (5) of the wall on W_0 , whose y^s -term carries $s(k_1+1)$, and the displayed choice of $h_{1,d}$ inherits the same slip. Only that display is affected: the inductive choice for $p \geq 1$ uses the correct form, and we use it throughout.

1.4. Method and verification. All algebraic and numerical claims are verified by exact polynomial and rational arithmetic in the accompanying programs (§8), which are available, together with this paper, at the public repository <https://github.com/bwuebben/open-fjrw-scattering>. An expository companion note on the central-charge threshold and the first-wall census is [Wue26]; the closed FJRW input for $x^5 + y^5$ is computed by three independent, mutually consistent routes, including a confirmation of every value through Witten’s r -spin class in `admcycles`, and the open topological recursion of [GKT22, Thm. 0.5] is verified in full against this data (§6).

2. THE WALL-CROSSING ALGEBRA OF CRITICAL GRAPHS

We recall the objects of [GKT22, §2.4, §4.3] in the notation used throughout. Internal markings carry twists (a, b) , $0 \leq a \leq r - 2$, $0 \leq b \leq s - 2$, and descendent exponents $d \geq 0$; boundary markings are of three types, σ_1 (twist $(r-2, 0)$), σ_2 (twist $(0, s-2)$), and the fully twisted σ_{12} (twist $(r-2, s-2)$). For a multiset J of internal twists with descendents $\mathbf{d}$, set $r(J) \equiv \sum_j a_j \pmod{r}$, $s(J) \equiv \sum_j b_j \pmod{s}$, and $m(J, \mathbf{d})$ as in (1).

Definition 2.1 ([GKT22, Not. 2.38]). Write $\sum_j a_j = r(J) + \ell_1 r$, $\sum_j b_j = s(J) + \ell_2 s$ and $N = N(J, \mathbf{d}) = \ell_1 + \ell_2 - |J| + 1 + \sum_j d_j$. The *balanced disk types* over $(J, \mathbf{d})$ are $\Gamma_{J,p}$, $0 \leq p \leq N$, with boundary degrees $(k_1, k_2) = (r(J) + pr, s(J) + (N-p)s)$ and one σ_{12} -point; the *critical*

graphs are $\Lambda_{J,p}$, $1 \leq p \leq N$, with boundary degrees $(r(J) + (p-1)r + 1, s(J) + (N-p)s + 1)$ and no σ_{12} -point. We refer to the set of balanced coefficients over a fixed $(J, \mathbf{d})$ as its *diagonal*. The balanced range is empty for $N < 0$, and the critical range is empty for $N < 1$. By [GKT22, Prop. 2.30], the diagonal is the set of lattice points of $sk_1 + rk_2 = m(J, \mathbf{d})$ obeying the congruences.

Proposition 2.2 (The primitive-form shift). *The Lie bidegree of the critical graph $\Lambda_{J,p}$, in the sense of (1), is its boundary degree minus $(1, 1)$ ([GKT22, Prop. 2.37]); equivalently, the wall element attached to Λ is the Hamiltonian vector field, for the primitive form $dx \wedge dy$, of the boundary monomial $x^{\# \sigma_1} y^{\# \sigma_2}$ of the critical disk. With $g_m = z^m(m_2 x \partial_x - m_1 y \partial_y)$ the tropical-vertex generator (Hamiltonian for $d \log x \wedge d \log y$), one has $X_{k_1, k_2} = (xy)^{-1} g_{(k_1+1, k_2+1)}$: the algebra is graded by an integral-affine lattice, the origin shifted by the primitive form, and the wall-crossings are the rational-power tropical vertex maps of [GKT22, Lem. 4.20]. The shift is the removal of the σ_{12} -insertion.*

Proof. The first statement is [GKT22, Prop. 2.37(b)] read against Definition 2.1. The dictionary $X_H^{\text{tor}} = xy X_H^{\text{aff}}$ follows from $\omega_{\text{aff}} = xy \omega_{\text{tor}}$, and the explicit wall maps are [GKT22, Lem. 4.20]. $\square$

Proof of Theorem 1.1. The constraint (1) reads $sk_1 + rk_2 = w(J, \mathbf{d})$ with $k_1, k_2 \geq 0$: for $w < 0$ there are no solutions, for $w = 0$ only $(0, 0)$, and for $w > 0$ the N solutions of Definition 2.1. A twist (a, b) contributes $sa + rb - rs \leq s(r-2) + r(s-2) - rs = rs(\hat{c}_W - 1)$ to the weight, with equality at the maximal twist; descendants contribute $rs d_j > 0$. Hence positive-weight $d = 0$ monomials exist if and only if $\hat{c}_W > 1$, and for $\hat{c}_W = 1$ every $d = 0$ monomial has $w \leq 0$ with equality exactly on products of marginal twists; whenever such an equality monomial is wall-bearing, its unique direction is the torus one. For the existence of walls (not merely of positive-weight monomials) when $\hat{c}_W > 1$: take m interior markings of maximal twist $(r-2, s-2)$; then $N = m - \lceil 2m/r \rceil - \lceil 2m/s \rceil + 1 \geq 1$ once $m(1 - \frac{2}{r} - \frac{2}{s}) \geq 2$, while the weight is $mrs(\hat{c}_W - 1) > 0$, so the resulting critical graphs carry wall summands; this is the maximal-twist construction of [Wue26]. At $\hat{c}_W = 1$, those analogous multisets for which $N \geq 1$ are critical and marginal, realizing the torus wall-crossing; for $\hat{c}_W < 1$, $\ell_1 + \ell_2 \leq m\hat{c}_W < m$ forces $N \leq 0$: no primary critical graphs at all. The qualifier in the theorem is necessary: acting on a lower divisor coefficient multiplies deformation monomials, so nonlinear transport can contribute to a nonpositive-weight target even though no generator is supported there. $\square$

Remark 2.3 (The marginal torus direction). At a wall-bearing marginal monomial ($w = 0$ and $N \geq 1$; necessarily $r(J) = s(J) = 0$ and $N = 1$) the critical graph exists (for $x^4 + y^4$ it is precisely the disk of [GKT26, Fig. 9, Ex. 4.9]), but its Lie bidegree is $(0, 0)$, the direction $x\partial_x - y\partial_y$ excluded from $\mathfrak{g}^{r,s}$ in [GKT22, (4.15), (4.18)] for nilpotency of the (x, y) -filtration. The exclusion is an artifact: with coefficients in the ideal generated by the deformation variables, the exponential converges adically, and the proofs of [GKT22, Thm. 5.8] (whose jump sets $Q_0(A, \mathbf{d})$ impose only (1), admitting $(0, 0)$) and of [GKT22, Cor. 5.13] (whose realizability step moves the two adjacent balanced coefficients by $h \cdot ((-1)^{k_2} s k_1(\Lambda), (-1)^{k_2+1} r k_2(\Lambda))$, at a marginal diagonal exactly the antisymmetric torus motion) use the torus summands. We therefore work with the enlarged algebra

$$\widetilde{\mathfrak{g}}^{r,s} = \mathfrak{g}^{r,s} \oplus \bigoplus_{\substack{\emptyset \neq J, \ w(J, \mathbf{d})=0, \ N(J, \mathbf{d}) \geq 1 \\ r(J)=s(J)=0}} \mathbb{Q} u_{J, \mathbf{d}} (x\partial_x - y\partial_y),$$

whose new summands are abelian among themselves and act diagonally on the graded pieces. This is a Lie algebra: for $D = X_{0,0}$, $[uD, vX_{a,b}] = (a-b)uvX_{a,b}$ and $[uD, vD] = 0$; since a

marginal coefficient u has weight and residues zero, uv satisfies the same critical-generator constraints as v . In the primary algebra nothing is added when $\hat{c}_W < 1$; for the parabolic pairs the primary group is purely torus, matching the elliptic wall-crossing group of Maher [Mah24, §5.2.2]; for $\hat{c}_W > 1$ the torus summands sit over the marginal submonomials. All statements below are over $\tilde{\mathfrak{g}}^{r,s}$ and its group $\tilde{G}^{r,s} = \exp(\tilde{\mathfrak{g}}^{r,s})$; the next lemma supplies the torsor properties over the enlargement.

Lemma 2.4 (The enlarged torsor). *The action of $\tilde{G}^{r,s}$ on potentials preserves $\Omega_{r,s}(I)$ and is free and transitive on it, so that $\Omega_{r,s}(I)$ is a torsor in the strict sense, and the jump automorphisms of [GKT22, Thm. 5.8] lie in $\tilde{G}^{r,s}$.*

Proof. Membership of the jumps: the jump sets $Q_0(A, \mathbf{d})$ in the proof of [GKT22, Thm. 5.8] impose only (1), so the jump elements are sums of pro-nilpotent and marginal torus summands.

Preservation and transitivity can be proved without inferring a full exponential action from a single leading coefficient change. The integration-by-parts argument of [GKT22, Thm. 4.26, Step 1], which proves that the group preserves $\Upsilon_{r,s}(I)$, uses the dimension constraint and the divergence-free Hamiltonian form of the generators. It applies unchanged to $X_{0,0} = x\partial_x - y\partial_y$; its coefficient has positive deformation degree, so the exponential converges in the deformation filtration. It also preserves the chamber-index normalization: for a primary singleton, residue zero forces twist $(0,0)$ and hence weight $-rs$, so a primary marginal torus monomial has $|J| \geq 2$ and acts trivially on the zero- and one-marking normalized terms; a descendent singleton torus monomial does not alter their $\mathbf{d} = 0$ normalization (the normalizations proper, [GKT22, Def. 3.46(1)], constrain only $\nu_{\Gamma,0}$; the remaining conditions there are of $\mathcal{A}$ -form and are preserved by the integration-by-parts argument). The triangular construction of [GKT22, Thm. 4.26, Step 3] then proves transitivity. For the new degree-zero vector field, refine its weighted-degree induction by deformation divisibility: a correction $u_{J,\mathbf{d}}X_{0,0}$ changes its target diagonal at leading order and only proper multiples thereafter. This secondary induction terminates modulo every finite divisor-closed coefficient cutoff and passes to the coefficientwise completion. The only new case is a marginal diagonal: it has two balanced coefficients and its allowed difference is exactly the one-dimensional antisymmetric direction $re_1 - se_0 = X_{0,0}(W_0)$: a marginal diagonal is NS-visible ($r(J) = s(J) = 0$), so its single $\mathcal{A}$ -condition is one nontrivial linear condition on the two coefficients, and the torus vector spans its kernel (the period computation in the proof of Theorem 1.4); hence the same induction closes. Finally $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ by [GKT22, Cor. 5.13]; hence the enlarged action preserves and is transitive on the realizable indices.

Freeness: let $g = \exp(v)$ with $v \neq 0$, and let u be minimal in the divisibility order among the monomials in the support of v . The change of the u -diagonal of any potential is exactly $v_u(W_0)$, $W_0 = x^r + y^s$: contributions of other summands, or of higher exponential terms, carry monomials properly containing a support monomial, which cannot equal u by minimality. And $v_u(W_0) \neq 0$. Indeed, on the diagonal belonging to u , the N pro-nilpotent generators act by the adjacent vectors $a_p e_p - b_p e_{p-1}$ with $a_p, b_p > 0$. These vectors are linearly independent: reading a relation successively in the $e_N, e_{N-1}, \dots, e_1$ coordinates kills all its coefficients. On a marginal diagonal there is instead the single nonzero torus vector $re_1 - se_0$. Thus no nonzero linear combination of the summands in v_u can cancel. (Individually, a pro-nilpotent summand contributes $r(k_2+1)x^{k_1+r}y^{k_2} - s(k_1+1)x^{k_1}y^{k_2+s}$, while a torus summand contributes $r x^r - s y^s$.) So g moves every potential, which is freeness and not merely faithfulness. The distinction matters downstream: uniqueness of the transition element, used in Theorem 1.2(i), needs trivial stabilizers, and faithfulness alone does not supply them for a nonabelian group. The argument of [GKT22, Thm. 4.26, Step 2] likewise establishes freeness, though it is stated there as faithfulness. All uses below are through this lemma. $\square$

3. THE SCATTERING PRESENTATION

Throughout, a *normal-form family* is a family of simple canonical homotopies supplied by [GKT22, Lem. 3.39]: for each subset A of the internal labels there is one time $t_A \in (0, 1)$, all times distinct, and every jump of a critical graph with internal marking set A occurs at t_A . The times are auxiliary choices (Remark 7.22).

Proof of Theorem 1.2(i). Let $(\mathfrak{s}_t)_{t \in [0,1]}$ be a family of simple canonical homotopies supplied by [GKT22, Lem. 3.39], with $\mathfrak{s}_1 = \mathfrak{s}_0$ and jump times $t_1 < \dots < t_M$ and jump elements $g_i \in \tilde{G}^{r,s}$, each a sum over the critical graphs with internal markings A_i , with coefficients, up to the explicit signs $(-1)^{|A_i|+k_2}$ (k_2 the Lie bidegree) of the display of g_i there, the zero counts $\Delta(\Lambda)$ of the homotopy on the critical stratum ([GKT22, proof of Thm. 5.8, eqns. (5.33)–(5.37)]). By that proof the potential $W^{\mathfrak{s}_t}$ is constant between jumps and transforms by g_i across t_i ; around the loop, $W^{\mathfrak{s}_0} = (g_M \circ \dots \circ g_1)(W^{\mathfrak{s}_0})$, and freeness (Lemma 2.4) gives $g_M \circ \dots \circ g_1 = \text{id}$. For two such paths with fixed endpoints, their total products g, g' satisfy $g'g^{-1}(W^{\mathfrak{s}_1}) = W^{\mathfrak{s}_1}$, hence $g' = g$. $\square$

Proof of Theorem 1.2(ii). Let $\mathfrak{m}$ be the deformation ideal and first work in the primary quotient

$$F^p \tilde{\mathfrak{g}}_0^{r,s} := \text{span}\{u_{J,0}X : |J| \geq p\}, \quad L_{<K} := \tilde{\mathfrak{g}}_0^{r,s} / F^K.$$

It is a finite-dimensional nilpotent Lie algebra and $[F^p, F^q] \subseteq F^{p+q}$ (here and below $F^\bullet$ also denotes the induced filtration on the quotient). For a rational boundary ray ρ let L_ρ be the span of the generators uX_{k_1,k_2} for which $(k_1+1, k_2+1) \in \rho$, placing a marginal generator $u(x\partial_x - y\partial_y)$ in $\rho = \mathbb{R}_{\geq 0}(1, 1)$. Then $L_{<K} = \bigoplus_\rho L_\rho$, with only finitely many summands.

Each L_ρ is abelian. Indeed,

$$[X_{k_1,k_2}, X_{k'_1,k'_2}] = \det((k'_1+1, k'_2+1), (k_1+1, k_2+1)) X_{k_1+k'_1, k_2+k'_2},$$

so the bracket vanishes when the shifted directions are parallel; on the $(1, 1)$ -ray the marginal vector field also commutes with every $X_{q,q}$. The deformation coefficient of a marginal generator has positive $\mathfrak{m}$ -degree, so it belongs to the same nilpotent filtration.

Fix any slope order. The multiplication map

$$\prod_{\rho} \exp(L_\rho) \longrightarrow \exp(L_{<K})$$

is bijective. Modulo F^2 this is the direct-sum decomposition above. Suppose the factors have been found through F^n . The logarithm of the residual in F^n/F^{n+1} decomposes uniquely by ρ ; insert these components into the corresponding factors. Commuting a degree- n correction past a previously chosen positive-degree term changes the product only in F^{n+1} . This proves existence by induction. At the first filtration degree where two ordered factorizations differ, the same direct-sum decomposition forces all differences to vanish, proving uniqueness. The construction is compatible as K varies.

For descendents, perform the same construction after quotienting to any finite divisor-closed set of deformation monomials. Its complement is upward closed, so this is a Lie-algebra quotient, and the compatible finite constructions give the asserted coefficientwise presentation. $\square$

Remark 3.1. [GKT22, Thm. 5.8] produces the total jump in a homotopy-time-ordered factorization, one factor per subset of internal markings, mixing wall directions; Lemma 3.39 of [GKT22] shows the jump times and their order are auxiliary choices. The boundary-ray presentation is a canonical reorganization by the *shifted* directions forced by Proposition 2.2, not by the additive

Lie rays (k_1, k_2) . The affine shift does not preserve rays; this filtered distinction is what makes the canonical diagram of Theorem 1.3 geometrically meaningful.

4. PER-DIAGONAL STRUCTURE: THE RECURSION TRANSVERSE TO THE WALLS

Two structural facts from [GKT22] organize this section. The open topological recursion factorizes through a cohomological-field-theory node splitting: in [GKT22, Lem. 6.4] the distinguished insertion splits off a closed extended invariant and continues as an open piece carrying the socle-dual twist $(a, b) \mapsto (r-2-a, s-2-b)$ of the Jacobi ring $\mathbb{C}[x, y]/(x^{r-1}, y^{s-1})$: the node is the residue pairing. And the Γ -weights in the invariant combinations $\mathcal{A}$ are the periods of $x^r + y^s$: $\int_0^\infty e^{-x^r} x^k dx = \frac{1}{r} \Gamma(\frac{1+k}{r})$, so the weight attached to a boundary profile is the period normalization of the primitive form. The good-basis period moments themselves are explicit:

Lemma 4.1 (Periods). *For the good-basis cycles Ξ_a of x^r ($\int_{\Xi_a} x^{a'} e^{x^r/h} = \delta_{aa'}$, $0 \leq a, a' \leq r-2$), the moments $P_a(p) = \int_{\Xi_a} x^p e^{x^r/h}$ satisfy $P_a(p) = \delta_{ap}$ for $p \leq r-2$, $P_a(r-1) = 0$, and $P_a(p) = -\frac{h(p-r+1)}{r} P_a(p-r)$.*

Proof. Total derivatives integrate to zero on Ξ_a : $0 = \int_{\Xi_a} \frac{d}{dx} (x^q e^{x^r/h}) = q P_a(q-1) + \frac{r}{h} P_a(q+r-1)$. $\square$

Proof of Theorem 1.4. The case $N = 0$. If $N = 0$, there is no wall direction and the nonzero $h = 1$ weight in $\mathcal{A}$ determines the sole coefficient modulo lower monomials, so the claim is immediate.

Wall-bearing diagonals. Fix now a wall-bearing diagonal $(J, \mathbf{d})$, of positive or marginal weight, with coefficients ν_p , $0 \leq p \leq N$, as in Definition 2.1. The wall element of $\Lambda_{J,p}$ acts on $W_0 = x^r + y^s$ by

$$(5) \quad u_{J,\mathbf{d}} X_{k_1,k_2}(W_0) = u_{J,\mathbf{d}} (r(k_2+1) x^{k_1+r} y^{k_2} - s(k_1+1) x^{k_1} y^{k_2+s}),$$

which lands on the two adjacent balanced points of the same diagonal: in the basis $e_0, \dots, e_N$ of coefficients, the wall p moves along the jump vector $w_p = a_p e_p - b_p e_{p-1}$ (not to be confused with the scalar Euler weight $w(J, \mathbf{d})$) with $a_p = r k_2(\Lambda_{J,p}) > 0$, $b_p = s k_1(\Lambda_{J,p}) > 0$; here and below $k_i(\Lambda)$ denotes the *boundary* degrees of Λ , which exceed the Lie bidegree (k_1, k_2) of the display by $(1, 1)$, so that $r(k_2+1) = r k_2(\Lambda)$ and $s(k_1+1) = s k_1(\Lambda)$, both positive. (These are exactly the geometric jump coefficients of [GKT22, (5.51)–(5.53)].) The w_p are triangular, hence independent, and each is annihilated by the period functional: by Lemma 4.1, $P_a(k_1+r) P_b(k_2) = -\frac{h(k_1+1)}{r} P_a(k_1) P_b(k_2)$ cancels against the second term of (5). *NS-visible diagonals.* On an NS-visible diagonal ($r(J) \leq r-2$ and $s(J) \leq s-2$) the period functional is nonzero on each e_p , so its kernel has dimension N and coincides with $\text{span}(w_p)$. By [GKT22, Thm. 4.9] the coefficient of $u_{J,\mathbf{d}}$ in $\int_{\Xi_{(r(J), s(J))}} e^{W^\nu/h} dx \wedge dy$ equals $(-1)^{|J|} (-h)^{-d(J,\mathbf{d})-2} \mathcal{A}(J, \mathbf{d}, \nu)$, whose $h = 1$ part (h the number of parts in the ordered-partition expansion of $\mathcal{A}$, [GKT22, Thm. 4.9]) is the period-weighted sum $\sum_p \omega_p \nu_p$ over the diagonal and whose $h \geq 2$ parts involve only strictly smaller monomials. The chamber-index conditions [GKT22, Def. 3.46] prescribe the value of $\mathcal{A}$.

Invisible diagonals. On invisible diagonals ($r(J) = r-1$ or $s(J) = s-1$) the good-basis periods vanish identically on the diagonal (Lemma 4.1), but the $h = 1$ weights of $\mathcal{A}$ remain nonzero there (their arguments $(1+K)/r$ are positive integers, where Γ does not vanish), and $\text{span}(w_p)$ lies in the kernel of the weighted functional modulo strictly smaller monomials, by the boundary-condition independence of $\mathcal{A}$ [GKT22, Thm. 5.4]; so the same one-linear-condition structure holds, imposed directly by the conditions, which by [GKT22, Cor. 5.13] characterize

$\Omega_{r,s}(I)$ completely. Ordering monomials by degree and descendents determines every invariant, up to the span of the w_p (the torsor directions), proving the theorem. The recursion [GKT22, Thm. 0.5] is the same system of conditions organized by descendent reduction, whence the last statement. $\square$

Remark 4.2 (The first computed tower). Call the family of diagonals over the powers of one deformation parameter its *tower*. For $x^5 + y^5$ the conditions produce, order by order over the $t_{3,3}$ -monomials (all verified by exact arithmetic; §8): the singleton normalization $\langle \tau_0^{(3,3)} \sigma_1^3 \sigma_2^3 \sigma_{12} \rangle = 1$; the first-wall relation (2) at $t_{3,3}^2$; the absolute value $\frac{1}{5}$ of (3) at $t_{3,3}^3$ (a diagonal with $N = 0$, hence no walls, and $r(J) = 4 = r - 1$, hence invisible to the good-basis periods, so the condition that determines it is not a period condition); and at $t_{3,3}^4$ a relation quadratic in the first-wall invariants, giving $\nu_{(2,7)}^{\min} = -\frac{12}{25}$ in the normal form of Theorem 1.3. The descendent seed at $d = 1$ gives $\langle \tau_1^{(3,3)} \sigma_1^8 \sigma_2^3 \sigma_{12} \rangle + \langle \tau_1^{(3,3)} \sigma_1^3 \sigma_2^8 \sigma_{12} \rangle = -\frac{5}{4}$.

Remark 4.3 (The seed values, and [GKT26, Ex. 5.5]). Imposing [GKT22, Thm. 0.3] on the oscillatory expansion of the potential directly — the mechanism by which [GKT22, Thm. 0.2] is proved, so that no combinatorial convention enters — forces, at every two-insertion seed diagonal $\{(a_1, b_1), (a_2, b_2)\}$ with $a_1 + a_2 = r$, $b_1 + b_2 = s$, $\mathbf{d} = 0$, the relation (4): the diagonal supports the boundary profiles $(r, 0)$ and $(0, s)$ only, $d(J, 0) < 0$ so the mirror theorem admits no term there, and the linear and quadratic insertions contribute $\nu_{(r,0)}/r + \nu_{(0,s)}/s$ and $1/(rs)$ respectively (Lemma 4.1; verified symbolically for $r = s = 4, 5$ and $(r, s) = (3, 5), (4, 6), (3, 7)$, §8). At $r = s = 4$ this gives $-\frac{1}{4}$ for the sum considered in [GKT26, Ex. 5.5], where $-\frac{1}{8}$ is stated; the discrepancy traces to the $h = 2$ partition count in the evaluation of $\mathcal{A}$ there, which [GKT22, Thm. 4.9] fixes as the count of ordered partitions of the labeled marking set with $1/h!$. The same count is pinned independently by the proven [GKT22, Thm. 5.4]: at $r = s = 5$, $\mathcal{A}(\{(1, 1), (1, 1)\}, 0)$ equals the closed three-point value $\langle \tau_0^{(1,1)3} \rangle = 1$, which the labeled count reproduces and the halved count does not (§8). On the marginal seed diagonals the two coefficients in (4) are moved (antisymmetrically) only by the torus direction of Remark 2.3; the symmetric gauge value is $\nu_{(r,0)} = \nu_{(0,s)} = -\frac{1}{2r}$ at $r = s$. The discrepancy concerns only the worked example; no theorem of [GKT22] or [GKT26] is affected.

5. NORMAL FORMS AND THE CANONICAL DIAGRAM $\mathfrak{D}^{r,s}$

Proof of Theorem 1.3. Work in a primary quotient modulo F^K , or more generally on a finite divisor-closed coefficient set as in Theorem 1.2(ii). For existence and uniqueness of $\nu^{\min}$, order its monomials by total degree. At each step, on each wall-bearing diagonal the span of the wall vectors $w_p = a_p e_p - b_p e_{p-1}$ ($a_p, b_p > 0$) is complementary to $\mathbb{Q}e_0$: a combination $\sum c_p w_p \in \mathbb{Q}e_0$ forces, reading the $e_N, e_{N-1}, \dots$ components in turn, $c_N = \dots = c_1 = 0$. Hence there is a unique group correction at this order making all ν_p , $p \geq 1$, vanish, and it does not disturb lower orders; on marginal diagonals the torus direction plays the role of the single wall. The resulting index is enumerative because the group preserves $\Omega_{r,s}(I)$ (Lemma 2.4). Uniqueness follows from freeness (Lemma 2.4) plus the same triangularity: a nontrivial leading correction moves some $\nu_{p \geq 1}$. The construction for $\nu^{\max}$ is symmetric (complement of $\mathbb{Q}e_N$). The boundary-slope-ordered factorization of the unique $g^{\min \rightarrow \max}$ (Theorem 1.2(ii)) is then canonical, and its wall functions are computed order by order from the normal forms. Uniqueness makes these constructions compatible under restriction, giving the primary completion and the coefficientwise descendent extension. $\square$

Example 5.1 (The first walls of $\mathfrak{D}^{5,5}$). At order two in the deformation variables the three wall-carrying diagonals of $x^5 + y^5$ have balanced profiles $(0, 6), (5, 1); (6, 0), (1, 5);$ and $(1, 6), (6, 1)$

respectively, with invariant relations and normal forms (§8):

monomial	relation	normal forms
$t_{2,3}t_{3,3}$	$\frac{2}{5}\nu_{(0,6)} + \frac{1}{5}\nu_{(5,1)} = -\frac{2}{25}$	$\nu_{(0,6)}^{\min} = -\frac{1}{5}, \nu_{(5,1)}^{\max} = -\frac{2}{5}$
$t_{3,2}t_{3,3}$	the image under $x \leftrightarrow y$	by symmetry
$t_{3,3}^2$	$\frac{2}{5}(\nu_{(1,6)} + \nu_{(6,1)}) = -\frac{4}{25}$	$\nu_{(1,6)}^{\min} = \nu_{(6,1)}^{\max} = -\frac{2}{5}$

Solving $W^{\nu^{\max}} - W^{\nu^{\min}} = c \cdot u X_{k_1, k_2}(W_0)$ on each diagonal is an overdetermined system (two monomial components, one coefficient) whose consistency is forced by the relations above and constitutes a check of the whole mechanism; it yields the wall functions of Theorem 1.3. The wall $t_{3,3}^2 X_{1,1}$ is the first symmetric pro-nilpotent primary wall: only the sum (2) is invariant, the difference jumps.

6. THE CLOSED VERTEX DATA FOR $x^5 + y^5$, AND THE VERIFIED RECURSION

The determination of Theorem 1.4 consumes closed extended FJRW invariants. For the first symmetric hyperbolic Fermat pair $x^5 + y^5$ ($\hat{c}_W = \frac{6}{5} > 1$) we compute this input exactly.

Example 6.1 (Closed vertex weights for $x^5 + y^5$). By Sebastiani–Thom the state space and cohomological field theory of the Fermat sum factor as $A_4(x) \otimes A_4(y)$, the tensor square of the 5-spin (A_4) theory; a closed invariant $\langle \prod_i \tau_{d_i}^{(a_i, b_i)} \rangle$ is the tensor correlator $\int_{\overline{\mathcal{M}}_{0,n}} W^x(\mathbf{a}) W^y(\mathbf{b}) \psi^{\mathbf{d}}$ with W^x, W^y the two Witten classes, of degrees $\varepsilon_1 = (\sum a_i - 3)/5$ and $\varepsilon_2 = (\sum b_i - 3)/5$. We compute these exactly (primitive-form normalization, sign $(-1)^{\varepsilon_1 + \varepsilon_2}$) by three independent, mutually consistent routes:

- the single-variable A_4 invariants from the Saito primitive form (flat coordinates, three-point residues, string/dilaton/topological-recursion descendents), validated against the A_2 normalization $\langle (\tau_0^1)^4 \rangle = \frac{1}{3}$, WDVV for A_2, A_3, A_4 , and the $r = 2$ Witten–Kontsevich numbers;
- the factorizing tensor correlators ($\varepsilon_1 = 0$ or $\varepsilon_2 = 0$), where one Witten class is the topological unit;
- the fully mixed correlators with $\min(\varepsilon_1, \varepsilon_2) = 1$, by reconstructing the divisor $\mathcal{W} = \sum_U c_U \delta_U$ on $\overline{\mathcal{M}}_{0,n}$ from its node-splitting restrictions and pairing with the other class, using a self-contained genus-zero tautological intersection calculus (splitting recursion, $\delta_S^2 = -\psi' - \psi''$), verified against Witten–Kontsevich and the Petersen intersection form of $\overline{\mathcal{M}}_{0,5} = \text{dP}_5$; each reconstructed class passes the independent check $\langle \mathcal{W} \psi^{n-4} \rangle = \langle \tau_1^{a_i} \dots \rangle_{A_4}$.

Representative values (denominators 5^{n-3} ; the socle pairing is $\langle \tau_0^{(0,0)} \tau_0^{(0,0)} \tau_0^{(3,3)} \rangle = 1$):

$$\begin{aligned}
\langle (\tau_0^{(2,1)})^3 \tau_0^{(2,0)} \rangle &= -\frac{2}{5}, & \langle \tau_0^{(1,3)} \tau_0^{(1,0)} (\tau_0^{(3,0)})^2 \rangle &= -\frac{1}{5} & (\varepsilon_1, \varepsilon_2) &= (1, 0), \\
\langle (\tau_0^{(2,1)})^2 \tau_0^{(2,2)} (\tau_0^{(1,2)})^2 \rangle &= \frac{4}{25}, & \langle (\tau_0^{(2,2)})^3 (\tau_0^{(1,1)})^2 \rangle &= \frac{2}{25} & (1, 1), \\
\langle (\tau_0^{(2,3)})^2 \tau_0^{(1,3)} \tau_0^{(1,2)} (\tau_0^{(1,1)})^2 \rangle &= -\frac{4}{125}, & \langle (\tau_0^{(2,3)})^2 (\tau_0^{(1,3)})^3 \tau_0^{(1,2)} \tau_0^{(0,1)} \rangle &= \frac{6}{625} & (1, 2), (1, 3).
\end{aligned}$$

The smallest case with both classes of codimension two, $n = 7$, is computed by the codimension-two stratum intersection form: e.g. $\langle \tau_0^{(3,1)} \tau_0^{(2,1)} (\tau_0^{(2,2)})^3 \tau_0^{(1,2)} \tau_0^{(1,3)} \rangle = \frac{34}{625}$, invariant under relabelling. All values are verified by the exact computations of §8.

As independent evidence for the entire framework, the open topological recursion of [GKT22, Thm. 0.5] is confirmed numerically for $x^5 + y^5$: writing the canonical combinations $\mathcal{A}(A, \mathbf{d})$

as closed extended invariants ([GKT22, Thm. 5.4]) and evaluating both sides, an exhaustive check covers every instance with insertions in the narrow range, and a second, independent computation removes the restriction by computing every invariant, Neveu–Schwarz and extended (Ramond, $\text{tw} = -1$) alike, through Witten’s r -spin (Chiodo) class in SageMath’s `admcycles`: the recursion holds in all 72 instances with $|J| = 2$ and all 105 with $|J| = 3$ (§8). As a by-product, the `admcycles` route independently reproduces every closed invariant of Example 6.1, including $\frac{34}{625}$ and the descendent values.

7. THE GEOMETRIC PICTURE

The results above define the canonical diagram $\mathfrak{D}^{r,s}$ and show that its consistency structure determines the invariants. What they do not provide is a *position space*: an intrinsic domain in which the walls are rays and the invariants are computed by broken lines, as for the Fano Landau–Ginzburg models of [Gro10, CPS24]. This section builds as much of that picture as is a theorem and isolates what is not.

Three words of warning on vocabulary. We reserve *base* for the base $B\Gamma$ of a graded graph [GKT22, Def. 3.5], an unrelated notion carrying the inductive structure of the boundary conditions, and call the domain below the *direction sector*; and *consistency* here always means the path-independence of Theorem 1.2(i) in boundary-condition space. The product $g^{\min \rightarrow \max}$ is the transport of a boundary-to-boundary sweep of the sector, whose two ends carry $\nu^{\min}$ and $\nu^{\max}$; it is not loop holonomy (Theorem 7.31). Likewise *chamber index* is [GKT22]’s name for a system of invariants and does not refer to a chamber of the direction sector.

The order below puts what is proved first. §7.1 places the walls, §7.2 proves the transport representation, its ordered-chain inversion and its corrected Euler-cascade class, and the sense in which the theory exceeds its own tropicalization; §7.3 realizes every finite coefficient window by staged homotopies, makes the realizations coefficientwise compatible, and proves the exact obstruction that forces the staging; and §7.4 proves the projective coherent-sector theorem and the resulting position-space obstruction.

7.1. The sector and its rays.

Definition 7.1. The *boundary-charge lattice* is $N_\partial = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$, where e_i records one boundary insertion of type σ_i . The *direction sector* is its effective cone $C_\partial = \mathbb{R}_{\geq 0}e_1 + \mathbb{R}_{\geq 0}e_2$, with the $(1, 1)$ -shifted integral-affine structure of Proposition 2.2; its two sides are the rays of the two boundary types. A wall of $\mathfrak{D}^{r,s}$ is placed on the ray $\mathbb{R}_{\geq 0}(k_1+1, k_2+1)$ of its critical graph’s boundary monomial, and a chamber of a finite primary truncation is a connected component of the complement of its rays. For the completed descendent diagram, a *slope cut* is a direction together with a choice of side when the direction itself carries a wall. The *base cut* is the side of the sector along the σ_1 -axis ray, and the transport $\Psi(q)$ to a cut q is the slope-ordered product of the wall factors between the base cut and q , defined on each finite divisor-closed coefficient set and then coefficientwise; only the composites $\Psi(q)\Psi(q')^{-1}$ enter below, so the choice of base is a convention. For $u, v \in \mathbb{Z}^2$ write $\text{cross}(u, v) = u_1v_2 - u_2v_1$.

Thus the quadrant, its lattice, and its axes come from the open FJRW boundary data rather than from an auxiliary drawing. Only its projectivization $\mathbb{P}(C_\partial) \cong [0, \infty]$, the ordered slope space, is used by transport. The placement is not a convention but forced, and the same computation shows why the theory looks tropical at all: the $(1, 1)$ -shift of Proposition 2.2, which is nothing but the removal of the σ_{12} -insertion, is exactly what makes bending along a transported monomial monotone in slope, hence what makes broken lines behave like broken lines here (Theorem 7.5).

Proposition 7.2 (Forced placement). *The wall element X_{k_1, k_2} fixes exactly the monomials proportional to its boundary-monomial direction (k_1+1, k_2+1) . Hence in any geometric realization the wall of a critical graph must lie on the ray of its boundary monomial (the $(1, 1)$ -shifted ray), and a seed monomial is supported on the ray of its own exponent. The direction sector of Definition 7.1 is thus forced, it sides the rays of the two boundary types; a full loop would have holonomy $g^{\min \rightarrow \max} \neq \text{id}$.*

Proof. $X_{k_1, k_2}(x^a y^b) = ((k_2+1)a - (k_1+1)b)x^{a+k_1}y^{b+k_2}$, and the coefficient is the determinant of (a, b) against (k_1+1, k_2+1) , vanishing precisely on the ray. The holonomy statement is Theorem 1.3. $\square$

Proposition 7.3 (Interlacing and extreme support). *On every diagonal the balanced monomials and the walls interlace in slope (writing bal_p for the balanced monomial of $\Gamma_{J,p}$), $\text{bal}_N < \Lambda_N < \dots < \Lambda_1 < \text{bal}_0$, and the normal forms of Theorem 1.3 are the extreme-chamber support conditions in every finite primary truncation: near one axis only the monomial at the far end of each diagonal survives.*

Proof. Immediate from the formulas of Definition 2.1: the wall Λ_p has boundary degree between those of bal_p and bal_{p-1} in slope; the support statement is the normal-form construction read through the interlacing. $\square$

7.2. The transport representation.

Definition 7.4. The *anchor cut* q_m of a potential monomial m is the one-sided cut through its own direction, on the side toward the $(1, 1)$ -direction. If m is off every wall in a finite truncation, this is represented by the chamber containing its direction; for m on the $(1, 1)$ -direction either side gives the same coefficient, and the exponent $(0, 0)$ is anchored at the base cut. Its *anchored value* is its coefficient at q_m . For a candidate assignment of seeds, $T(q) := \sum_m \Psi(q)\Psi(q_m)^{-1}(\text{seed}_m)$ denotes the resulting transport potential.

Every bend along a transported monomial adds $\beta - (1, 1)$ to the exponent, β the boundary direction on the crossed ray. Throughout, seed and target are taken off the crossed rays: a seed on a bend ray has vanishing bend coefficient, and an on-ray target uses the one-sided anchor cut toward the $(1, 1)$ -direction, the convention of Theorem 1.5. Call a bending contribution to a coefficient *foreign* if it originates from a seed other than the coefficient's own.

Theorem 7.5 (No backwards bending). *Along a path with b bend factors from a seed with exponent m_σ into a monomial m ,*

$$b \text{ cross}(m, (1, 1)) = \text{cross}(m, m_\sigma) + \sum_f \text{cross}(m, \beta_f).$$

Consequently no foreign bending reaches a monomial's anchor cut from the far side of the monomial's own direction, the side away from the $(1, 1)$ -direction: a monomial below the $(1, 1)$ -direction receives foreign bending only along paths descending in slope, one above it only along paths ascending, and monomials on the $(1, 1)$ -direction receive no foreign contribution at all.

Proof. Summing the per-bend exponent motion gives $m + b(1, 1) = m_\sigma + \sum_f \beta_f$; the displayed identity is its cross product with m . Transport is monotone, so every crossed ray lies strictly between the seed's direction and m 's. For a path from below, seed direction and all crossed rays are strictly shallower than m , so the right side is negative, forcing $\text{cross}(m, (1, 1)) < 0$, i.e. m strictly above the $(1, 1)$ -direction; from above it is positive, forcing m strictly below. A monomial on the $(1, 1)$ -direction, where the cross product vanishes, excludes both. $\square$

The primitive-form shift is thus precisely the monotone-bending mechanism.

Throughout, *seed* means the initial datum of a broken line in the sense of [Gro10, CPS24] and carries no cluster-theoretic meaning.

Lemma 7.6 (Reduction). *For the anchored-value assignment, $T = W$ at every cut if and only if, for every coefficient, the foreign bending contributions cancel at its anchor cut.*

Proof. Wall automorphisms are additive, so $W - T$ transports as a family; its coefficients on any divisibility-minimal nonzero monomial are cut-independent (all bending contributions come from proper divisors, where $W - T$ vanishes), and evaluating each coefficient at its anchor cut gives zero exactly under the stated cancellation. $\square$

Theorem 7.7 (Linear order; singleton towers). *At linear order in a diagonal's own wall functions applied to the axis seeds x^r, y^s , the cancellation of Lemma 7.6 holds on every diagonal of every $(r, s, J, \mathbf{d})$: transporting into the cut at bal_p from below crosses exactly the walls Λ_q , $q \geq p+1$, whose x^r -bends land on bal_q , never on the target coefficient, while from above the crossed walls' y^s -bends land on bal_{q-1} , $q \leq p$, again off the target: the interlacing pattern is the no-backwards-bending property of broken lines in [CPS24]. Bends of divisor seeds at foreign walls are outside this statement; Example 7.18 shows they can arrive. For singleton diagonals (one internal marking, any descendent) no cross-diagonal contribution exists, the linearization is exact, and the transport representation holds with the anchored-value assignment on every such tower, with wall functions the chain $c_{q+1} = c_q a_q / b_{q+1}$ closed by the $\mathcal{A}$ -condition.*

Proof. The crossing sets and landing indices are those of Proposition 7.3 together with the landing identities: by (5), $X_{\Lambda_q}(x^r)$ is a nonzero rational multiple of the monomial bal_q and $X_{\Lambda_q}(y^s)$ of bal_{q-1} . For a singleton monomial no proper divisor exists, so the only contributions are the linear ones. $\square$

Two computed examples below (Examples 7.17 and 7.18) calibrate what the seeds are.

Theorem 7.8 (Transport representation; = Theorem 1.5). *With the anchor assignment of Definition 7.4, the difference between the potential's coefficient vector on a monomial and the transported contributions of the seeds of its proper divisors is independent of the cut. Defining the seed system by these constants, every cut potential is the transport*

$$W(q) = \sum_m \Psi(q) \Psi(q_m)^{-1} (\text{seed}_m)$$

of the seeds, and the seed system is the unique one with this property.

Proof of Theorem 1.5. Induct on the divisibility order of deformation monomials; the base is $W_0 = x^r + y^s$, whose two coefficients are seeds with value 1 in every cut. Fix a monomial $u = u_{J, \mathbf{d}}$ and assume the seeds are defined, with the transport identity holding, on every proper divisor of u . Write D for the diagonal of u , so that $u_D = u$, and E for the diagonal of a proper divisor u_E of u . Let $F(q)$ denote the u -component of the sum of the transported lower seeds.

Step 1: the jump formula. Across any wall $\theta = \exp(\eta)$ the coefficients on D jump by

$$\Delta_\theta(W|_D) = \sum_{j \geq 1} \frac{1}{j!} \sum_{\substack{\eta_1, \dots, \eta_j \in \text{supp}(\eta) \\ u_E u_{\eta_1} \cdots u_{\eta_j} = u_D}} [\eta_1 \cdots \eta_j (W|_E)]_D,$$

a universal expression in the wall function and the coefficients of W on proper divisors E of D and on W_0 : any other source overshoots u , and D 's own walls act on D 's coefficients only through W_0 (their monomial is u itself), so the jump is exactly linear there.

Step 2: cancellation, and chamber-independence of the residual. The jump of F across the same wall is the same universal expression evaluated on the transported-lower-seed sum, whose divisor coefficients equal W 's by the induction hypothesis and whose W_0 -coefficients are the same constants 1. Hence the two jumps cancel in the difference and $V := W(q)|_D - F(q)$ is independent of q .

Step 3: the seed at D , and uniqueness. A u -term transports with unchanged u -component (every wall action on it overshoots u), so placing $\text{seed}(u) := V$ at the assigned anchor cuts gives $T|_D = V + F = W|_D$ at every cut (all sums are coefficientwise finite: only finitely many divisor seeds and bend factors contribute to a fixed monomial), completing the induction. Uniqueness is immediate: V is determined by W and the lower seeds. Anchor assignments affect only the higher-order terms of the transports, hence the seed values at higher orders, never the existence or uniqueness of the decomposition; the $(1, 1)$ -side assignment is the canonical choice, being the one compatible with Theorem 7.5. $\square$

Lemma 7.9 (Ordered bend expansion). *Work in a finite divisor-closed coefficient quotient and write $z^a = x^{a_1}y^{a_2}$. Suppose a transport path crosses, in chronological order,*

$$\Theta_j = \exp(\epsilon_j \eta_j), \quad \eta_j = \sum_{\lambda \in L_j} c_\lambda u_\lambda X_{k_\lambda}, \quad \epsilon_j \in \{1, -1\}.$$

For nonnegative integers n_j and ordered choices $\lambda_{j,1}, \dots, \lambda_{j,n_j} \in L_j$, read the decorations first by increasing j and put $a^h = a^{h-1} + k_{\lambda_h}$, starting from $a^0 = a$. Then

$$(6) \quad (\Theta_m \circ \dots \circ \Theta_1)(u_E z^a) = \sum_{\gamma} \frac{\prod_h \epsilon_{j(h)} c_{\lambda_h} \det(a^{h-1}, k_{\lambda_h} + (1, 1))}{\prod_j n_j!} u_E \prod_h u_{\lambda_h} z^{a^{|\gamma|}},$$

the finite sum over the surviving decorated bend chains γ .

Proof. Expand each exponential in powers of η_j and distribute each power over its summands. Successive applications use

$$X_k(z^a) = ((k_2 + 1)a_1 - (k_1 + 1)a_2)z^{a+k} = \det(a, k + (1, 1))z^{a+k}.$$

The exponential supplies $1/n_j!$, the path supplies the displayed factor order, and nilpotence makes the sum finite. Generators on one boundary ray commute, so repeated bends may equivalently be grouped as multisets with the same factorials. Notice that the expansion is by chains: transport is linear in its initial monomial. Rooted Lie trees arise only after an additional, noncanonical BCH expansion of the ray factors. $\square$

Index a potential slot by $i = (D, a)$, where D is a deformation monomial and a an exponent on its diagonal; let q_i be its anchor cut. If $j = (E, b)$, define the strictly divisibility-triangular transport kernel

$$(7) \quad K_{ij} = [u_D z^a] \Psi(q_i) \Psi(q_j)^{-1} (u_E z^b) \quad \text{when } E \mid D, E \neq D, \quad K_{ij} = 0 \quad \text{otherwise.}$$

Lemma 7.9 is an explicit formula for K_{ij} : retain the chains whose deformation labels multiply u_E to u_D and whose exponent increments carry b to a .

Theorem 7.10 (Triangular seed inversion). *On every finite divisor-closed coefficient set, let A_i be the coefficient of slot i at its anchor cut and let κ_i be its canonical seed. In vector form,*

$$(8) \quad A = (1 + K)\kappa, \quad \kappa = (1 + K)^{-1}A = \sum_{n \geq 0} (-K)^n A.$$

In particular,

$$(9) \quad \kappa_i = A_i + \sum_{n \geq 1} (-1)^n \sum_{i_0 < \dots < i_n = i} K_{i_n i_{n-1}} \cdots K_{i_1 i_0} A_{i_0},$$

where $<$ includes strict divisibility of the deformation monomials and the intermediate boundary exponents are summed. Thus every seed is a finite alternating sum of anchored open FJRW coefficients weighted by ordered broken-line chains.

Proof. Evaluate Theorem 1.5 at the anchor of i . The seed in slot i contributes κ_i , since a positive-degree wall acting on it overshoots its deformation monomial. A seed in $j < i$ contributes $K_{ij}\kappa_j$, and no other seed contributes. Hence $A_i = \kappa_i + \sum_{j < i} K_{ij}\kappa_j$. The kernel strictly increases deformation monomials and is nilpotent in the finite quotient, so its geometric series inverse is finite and gives (8)–(9). $\square$

The inversion has a zero-dimensional enumerative lift. Let $\mathcal{Z}_i^{\text{anc}}$ be the signed, automorphism-weighted zero-cycle of the balanced descendent Witten multisection at q_i , so $\deg \mathcal{Z}_i^{\text{anc}} = A_i$. For a term $c_\lambda u_\lambda X_{k_\lambda}$ in a ray logarithm, let $\mathcal{Z}_\lambda^{\text{crit}}$ be the signed weighted union of the zero loci of the corresponding critical-graph homotopies in the staged realization. Exact-support ray purity identifies its degree with c_λ .

For a decorated bend chain γ beginning at $i_0 = (E, b)$, define the rational oriented cascade cycle

$$(10) \quad \mathcal{C}_\gamma = \frac{\prod_h \epsilon_j(h) \det(a^{h-1}, k_{\lambda_h} + (1, 1))}{\prod_j n_j!} \left(\mathcal{Z}_{i_0}^{\text{anc}} \times \prod_h \mathcal{Z}_{\lambda_h}^{\text{crit}} \right).$$

Equations (5.36)–(5.37) of [GKT22] identify one such bend with the corresponding critical boundary zero locus: its zero count is the critical-homotopy count times the balanced count, multiplied by precisely the determinant/gluing factor in (10). Associativity of the assembling operation [GKT22, §2.3.2] allows these factors to be iterated. Expanding the kernels along a strict divisibility chain $i_0 < \dots < i_n = i$ therefore gives a finite cascade cycle $\mathcal{C}(i_0 < \dots < i_n = i)$ of degree $K_{i_n i_{n-1}} \cdots K_{i_1 i_0} A_{i_0}$.

Theorem 7.11 (Corrected relative-Euler cascade class). *The rational zero-cycle*

$$(11) \quad \mathcal{Z}_i^{\text{seed}} = \mathcal{Z}_i^{\text{anc}} + \sum_{n \geq 1} (-1)^n \sum_{i_0 < \dots < i_n = i} \mathcal{C}(i_0 < \dots < i_n = i)$$

is finite and satisfies

$$\deg \mathcal{Z}_i^{\text{seed}} = \kappa_i.$$

Its class in rational oriented zero-bordism is independent of the endpoint representatives, canonical homotopies, and transverse perturbations used in the staged realization.

Proof. Taking degrees in (11) gives the right side of (9), hence κ_i . Anchored degrees are coefficients of the canonical cut potentials, and the critical degrees are coefficients of the canonical ray factors; freeness and ray purity make both independent of the chosen representatives. Finally, rational oriented zero-bordism is classified by weighted degree, $\Omega_0^{\text{or}} \otimes \mathbb{Q} \cong \mathbb{Q}$, which proves independence of the class. $\square$

We next package the separate products in (11) into one geometric object. A *virtual orbifold with corners* below means a finite Kuranishi atlas with isotropy: its charts are finite-dimensional orbifolds-with-corners carrying relatively oriented obstruction bundles, and a codimension- c coordinate change stabilizes both the base and the obstruction bundle by the same c normal directions. A compatible transverse multisection defines a weighted branched virtual cycle

[MW17]. All charts used here are products and corners of the finite-dimensional GKT orbifolds; no additional analytic regularization is involved.

Lemma 7.12 (Polytope-relative canonical extension). *Let P be a compact smooth polytope with corners and let $\mathcal{G}$ be a finite construction-closed collection of graded graphs. Suppose compatible canonical multisection families are prescribed on*

$$(\partial P \times \mathcal{PM}_\Gamma) \cup (P \times \partial \mathcal{PM}_\Gamma)$$

and near $P \times \partial^+ M_\Gamma$, for every $\Gamma \in \mathcal{G}$. Assume that the graph-boundary values are assembled from the corresponding families on the base components and are strongly positive near the positive boundary. The families extend, relative to a smaller collar of the prescribed set, over $P \times \mathcal{PM}_\Gamma$. They may be chosen automorphism-invariant and transverse on every parameter face and graph stratum. Two extensions fixed on the prescribed set are joined by such an extension over $P \times [0, 1]$.

Proof. Repeat the induction on $\dim \mathcal{PM}_\Gamma$ in [GKT22, Lem. 3.39]. Once the lower graph families are fixed, assembling prescribes the section on $P \times \partial \mathcal{PM}_\Gamma$; associativity of assembling [GKT22, §2.3.2] makes the prescriptions agree at graph corners. Orbifold collars and the first three steps of the BCT extension argument used in that proof extend the data while preserving strong positivity. On the compact complement choose finitely many local multisections spanning the fibers. Generic small linear combinations are simultaneously transverse on the finitely many parameter faces and graph strata. Taking the finite automorphism orbit gives an invariant multisection. The same construction over $P \times [0, 1]$ proves the final assertion. $\square$

Lemma 7.13 (Iterated critical corner). *Let $\gamma = (\lambda_1, \dots, \lambda_m)$ be a labeled decorated bend history acting on an anchored balanced problem. There is a stable graph Γ_γ with m distinguished boundary edges and a distinguished balanced spine such that $\mathcal{PM}_{\Gamma_\gamma}$ is a codimension- m corner of the fully smoothed outer moduli space. Detaching the distinguished edges gives a finite map*

$$F_\gamma : \mathcal{PM}_{\Gamma_\gamma} \longrightarrow \mathcal{PM}^{\text{bal}} \times \prod_{h=1}^m \mathcal{PM}_{\lambda_h}^{\text{crit}}.$$

The descendent Witten bundle and assembled multisection on the corner are the pullbacks of the external sums on the right. These identifications do not depend on the order of detachment.

Proof. For $m = 1$ this is the critical boundary graph of [GKT22, Not. 5.6]: Definition 2.6 and Equation (2.7) there give the codimension-one closed stratum, while Proposition 2.14 and Observation 3.11 identify the bundle; canonicity gives the section identity used in (5.37). Inductively attach the next critical component to the rooted balanced component. The labeled realization gives disjoint labels to repeated insertions, so the graph remains stable. Proposition 2.4 of [GKT22] supplies the orbifold corner and one smoothing coordinate per edge. The assertion follows by iterating detachment; independence of order is precisely associativity of assembling. $\square$

For $j < i$, let $\mathfrak{T}_{ij}$ be the finite rational virtual zero-orbifold obtained by taking the union of the decorated bend charts in the transport from q_j to q_i , including the GKT gluing covers, signs, and symmetric quotients. Lemma 7.13 and [GKT22, (5.36)–(5.37)] give

$$(12) \quad \deg[\mathfrak{T}_{ij}]^{\text{vir}} = K_{ij}.$$

Theorem 7.14 (Compact virtual cascade realization). *Fix a finite divisor-closed coefficient set and a target slot i . There is one compact oriented virtual orbifold with corners $\overline{\mathfrak{C}}_i^{\text{bar}}$ of virtual*

dimension zero, one assembled obstruction atlas on it, and one coherent transverse multisection $\mathfrak{s}_i$ such that

$$[Z(\mathfrak{s}_i)]^{\text{vir}} = \mathcal{Z}_i^{\text{seed}}.$$

In particular its virtual degree is κ_i . Its virtual zero-bordism class is independent of all paths, collars, higher fillers, and transverse perturbations. These objects can be chosen compatibly under enlargement of the coefficient set.

Proof. We first verify the nontrivial composition face. For $i_0 < i_1 < i_2$, use Moore paths in the staged boundary-condition interval. The concatenated path through q_{i_1} and the direct path from q_{i_0} to q_{i_2} are homotopic relative to their endpoints. Pull back the GKT families over this homotopy, use compactified ordered configurations for the bend times, and apply Lemma 7.12. After the broken configuration faces are identified by Lemma 7.13, the zero locus is a compact virtual one-bordism with

$$(13) \quad \begin{aligned} \partial[\mathfrak{B}(i_0, i_1, i_2)]^{\text{vir}} &= [\mathfrak{T}_{i_2 i_1} \times \mathfrak{T}_{i_1 i_0} \times \mathcal{Z}_{i_0}^{\text{anc}}]^{\text{vir}} \\ &\quad - [\mu_2(\mathfrak{T}_{i_2 i_1}, \mathfrak{T}_{i_1 i_0}) \times \mathcal{Z}_{i_0}^{\text{anc}}]^{\text{vir}}. \end{aligned}$$

Here μ_2 concatenates the histories and forgets the waypoint; it retains the same critical and balanced factors. Thus (13) is a cycle-level boundary identity, not only the numerical equality $\deg \mu_2 = K_{i_2 i_1} K_{i_1 i_0}$.

The higher coherences are obtained inductively over the multiplihedra J_m , whose faces encode the compositions of consecutive subwords [MW10]. The lower families agree on intersecting faces because Moore-path concatenation and GKT assembling are associative. Lemma 7.12 fills $J_m \times \mathcal{PM}_\Gamma$ relative to its boundary. Consequently the finite directed system of transport correspondences and anchored cycles has a homotopy-coherent relative bar realization.

Its nondegenerate n -cells are

$$\coprod_{i_0 < \dots < i_n = i} \mathfrak{T}_{i_n i_{n-1}} \times \dots \times \mathfrak{T}_{i_1 i_0} \times \mathcal{Z}_{i_0}^{\text{anc}}.$$

Collapse faces compose adjacent transports, breaking faces are the multi-critical corners of Lemma 7.13, and the remaining external faces are relative faces with a nonzero normal section. On an n -cell, cross the product GKT chart with the cellular n -handle and add its n normal directions to the obstruction bundle. On a face this is the stabilization by the boundary coordinate. More explicitly, a breaking-face coordinate change is the span consisting of the finite labeled detaching cover and the closed corner embedding of [GKT22, Def. 2.6]; a collapse-face coordinate change is the collared parameter-face embedding. Proposition 2.14 and assembling identify the bundle data, and the smoothing coordinate identifies the tangent and obstruction normal quotients. Thus the spans satisfy the Kuranishi index condition and, by the higher face identities, form a finite atlas with isotropy [MW17]. The standard cellular Morse section has one zero of Morse index n and hence local sign $(-1)^n$. Keeping the original product multisections near these zeros and applying Lemma 7.12 elsewhere gives

$$[Z(\mathfrak{s}_i)]^{\text{vir}} = \mathcal{Z}_i^{\text{anc}} + \sum_{n \geq 1} (-1)^n \sum_{i_0 < \dots < i_n = i} [\mathfrak{T}_{i_n i_{n-1}} \times \dots \times \mathfrak{T}_{i_1 i_0} \times \mathcal{Z}_{i_0}^{\text{anc}}]^{\text{vir}},$$

which is (11).

It remains to check the rational weights. Permuting two virtual zero-dimensional critical factors changes the base orientation by $(-1)^{d_1 d_2}$ and the obstruction orientation by $(-1)^{r_1 r_2}$; since $d_a = r_a$, their product is 1. Thus the S_{n_j} -quotient for n_j same-ray bends is relatively orientation-preserving and contributes $1/n_j!$. The GKT boundary orientation and finite gluing

degree give the determinant factor in (10). Hence the displayed virtual cycle is the stated cycle itself. Its degree is κ_i by Theorem 7.10.

Strict divisibility bounds the bar length, so the finite realization is compact. Applying Lemma 7.12 after adding one parameter gives independence by a virtual one-bordism. Applying its relative form along the construction-closed exhaustion, with the old charts fixed, gives coefficientwise compatibility. $\square$

Remark 7.15 (Virtual versus ordinary presentation). Theorem 7.14 is the single cascade object naturally provided by the geometry. It is stronger than the formal zero-bordism class of Theorem 7.11: its cells are attached by the actual GKT detaching, smoothing, and continuation faces. It does not assert a further presentation by one smooth GKT orbifold and one fixed-rank ordinary orbibundle. Across an n -cell face, one cellular base direction and one obstruction direction disappear together; the Kuranishi stabilization records this canonically. A fixed-rank presentation would require a separate global thickening theorem and is not needed for the virtual Euler cycle.

The mechanism is corroborated computationally. At orders t^2 and t^3 the pro-nilpotent wall diagonals are verified in full: their coefficients, with their cut dependence and signs, are generated by the axis seeds x^r, y^s bending at the computed wall functions. In the pure- $t_{3,3}$ family the verification extends through t^4 , nonlinearly (§8): the ansatz $\exp(\frac{1}{50}t_{3,3}^2X_{1,1} + c_4t_{3,3}^4X_{2,2})$ carries $W^{\nu^{\min}}$ to $W^{\nu^{\max}}$ if and only if $c_4 = 0$, both order-4 coefficients pinning c_4 independently, and the double bend of the y^5 seed computes $\langle \tau_0^{(3,3)^4} \sigma_1^2 \sigma_2^7 \sigma_{12} \rangle^{\min} = -\frac{12}{25}$ from the wall function alone. The mixed $t_{2,3}$ – $t_{3,3}$ family through t^4 (the first with noncommuting walls) passes the same checks (§8), with order-4 wall coefficients $c_{(1,2)} = -\frac{1}{750}$ (with a contribution from the commutator $[\theta_{(1,2)}, \theta_{(1,1)}]$, the first scattering vertex of the theory) and $c_{(0,2)} = -\frac{1}{1250}$.

In the off-ray case of Example 7.18 the canonical seed of the extreme coefficient (7, 1) is $-\frac{10}{3}$: its normal-form zero is the superposition of the seed with a *far-side bend* (a surviving foreign contribution at the coefficient's anchor cut; Example 7.18), $0 = -\frac{10}{3} + \frac{10}{3}$ (with the naive anchored-value seed the transport misses W by the constant $-\frac{10}{3}$; with the canonical seed it agrees exactly in all four chambers of that finite family). On the foreign-free coefficients (those whose anchor cut receives no foreign bending) the seed coincides with the anchored value, zero at extremes. Every coefficient on the (1, 1)-direction is foreign-free (Theorem 7.5), and the exhaustive enumeration behind Example 7.18 verifies that all extreme coefficients of the two computed families are foreign-free as well: those coefficients are purely tropical. This establishes the final clauses of Theorem 1.5. Theorems 7.11 and 7.14 supply the zero-dimensional enumerative class of every seed and its single compact virtual cascade realization. The position-space reading of the direction sector remains open.

Observation 7.16. The open theory is strictly richer than its tropicalization. Broken lines generate exactly the part of the potential reachable by transport from the seeds; the seed system is input the diagram does not determine, and it is not zero. This has no analogue in the Fano case of [Gro10, CPS24], where every correction is disk-generated. The witness is Example 7.17: a middle seed of value $-\frac{25}{16}$ that no bending can produce.

Example 7.17 (The non-tropical part; middle seeds). Two things this example shows: bending cannot generate everything, and seeds are not confined to the coefficients that carry no wall. For the first, the invariant $\langle \tau_0^{(3,3)^3} \sigma_1^4 \sigma_2^4 \sigma_{12} \rangle = \frac{1}{5}$ of (3) has no bending source (t -monomials multiply along a broken line, and the unique candidate wall fixes the $t_{3,3}$ -seed exactly), so such coefficients are seeds. For the second, take the $N = 2$ singleton tower $t_{3,3,d=2}$ of $x^5 + y^5$,

the exact solution of Theorem 7.7 has equal wall coefficients $c_1 = c_2 = \frac{5}{144}$, extreme value $\langle \tau_2^{(3,3)} \sigma_1^3 \sigma_2^{13} \sigma_{12} \rangle = \frac{25}{36}$, and *middle seed* $\langle \tau_2^{(3,3)} \sigma_1^8 \sigma_2^8 \sigma_{12} \rangle^{\text{mid}} = -\frac{25}{16}$, the $\mathcal{A}$ -condition holding in all three chambers.

Example 7.18 (Far-side bends). Theorem 7.5 excludes foreign bending only from beyond the target's own direction; arrivals from the $(1, 1)$ -side survive it. We call a surviving foreign contribution at a target's anchor cut a *far-side bend*, after the path shape: the seed lies on the far side of the crossed wall from the target's cut. Characteristically these are near-axis descendent walls bending seeds from the linear, flat-coordinate part of the deformation. Exhaustive enumeration shows the two computed families of $x^5 + y^5$ admit no such configurations (their verified transport identities are fully explained by Theorems 7.5 and 7.7), but composite diagonals with descendents do. The point of the two smallest cases is that the naive rule fails and the canonical seed is what repairs it: in the off-ray case the anchored value is 0 while the seed is $-\frac{10}{3}$, the difference being exactly a far-side bend. In detail, $J = \{(0, 0, 2), (1, 1, 0)\}$ and the off-ray $J = \{(0, 0, 2), (2, 1, 0)\}$ are solved completely (§8): the unit-twist tower $t_{0,0,2}$ carries nonzero wall coefficients $\frac{5}{6}$ on both of its near-axis rays, the composite diagonals have forced own wall coefficients 1, and the $\mathcal{A}$ -conditions hold in every chamber of the finite family. In the off-ray case the far-side bend contributes $\frac{10}{3}$ to the *extreme* coefficient $(7, 1)$ in its anchor cut: the anchored value (0, the normal form) is not the seed, and the correct object is canonical.

7.3. Realization by staged homotopies.

Proposition 7.19 (Descendent walls accumulate; the primary diagram is locally finite). *Fix $r, s \geq 2$ and an interior twist (a, b) . Let d range over $\mathbb{Z}_{\geq 1}$, so that $t_{a,b,d}$ runs over deformation parameters of total degree one. The rays of the associated critical graphs are exactly*

$$R(a, b) = \{(a + jr + 1, b + \kappa s + 1) : j, \kappa \in \mathbb{Z}_{\geq 0}\}, \quad d = j + \kappa + 1,$$

their wall functions in $\mathfrak{D}^{r,s}$ are all nonzero, and their slopes are dense in $(0, \infty)$. Hence the walls of $\mathfrak{D}^{r,s}$ accumulate in every subsector of the direction sector already at total degree one in the deformation parameters, for every (r, s) , the simple singularities included; those carry no primary wall at all (Theorem 1.1). By contrast, write $\mathfrak{D}_0^{r,s}$ for the primary diagram, the walls with $\mathbf{d} = \mathbf{0}$, and $\mathfrak{m}$ for the ideal generated by the deformation parameters. Then $\mathfrak{D}_0^{r,s}$ has finitely many walls modulo $\mathfrak{m}^n$ for every n , and its chambers are open angular sectors.

Proof. For a singleton $J = \{(a, b)\}$ one has $\ell_1 = \ell_2 = 0$ and $N = d$, so by Definition 2.1 the critical graphs $\Lambda_{J,p}$, $1 \leq p \leq d$, have boundary degrees $(a + (p-1)r + 1, b + (d-p)s + 1)$; putting $j = p-1$, $\kappa = d-p$ gives $j + \kappa = d-1$ and the stated set, every element of which is attained by taking $d = j + \kappa + 1$. For density, fix $t > 0$ and let j be large enough that $t(a + jr + 1) > 1 + b$; with $\kappa = \lceil (t(a + jr + 1) - 1 - b)/s \rceil$ one has $\kappa s \in [t(a + jr + 1) - 1 - b, t(a + jr + 1) - 1 - b + s)$, whence

$$\left| \frac{b + \kappa s + 1}{a + jr + 1} - t \right| < \frac{s}{a + jr + 1} \xrightarrow{j \rightarrow \infty} 0.$$

For the wall functions, note that on a singleton diagonal the chamber-index condition is $\mathcal{A}(J, \mathbf{d}, \nu) = (-1)^d \neq 0$, whose $h = 1$ part is $\sum_p \omega_p \nu_p$ with all $\omega_p > 0$, so $\nu_N^{\max} \neq 0$; solving $W^{\nu^{\max}} - W^{\nu^{\min}} = \sum_p c_p w_p$ with $w_p = a_p e_p - b_p e_{p-1}$, $a_p, b_p > 0$, gives $c_N = \nu_N^{\max}/a_N \neq 0$ and $c_p = c_{p+1} b_{p+1}/a_p \neq 0$ for every p , and no cancellation between distinct rays is possible at total degree one, where the monomials are distinct. For the primary statement, modulo $\mathfrak{m}^n$ only monomials $u_{J,0}$ with $|J| < n$ occur, and for each such $|J|$ the twists range over the finite Neveu–Schwarz box, so only finitely many diagonals, hence finitely many walls, contribute; finitely many rays through the origin cut the sector into open angular sectors. $\square$

In the statement and proof of the next theorem, call a jump, stage, or degree *visible* if its class modulo the stated quotient is nonzero; this sense is unrelated to the NS-visibility of diagonals.

Theorem 7.20 (Finite-support and pro-geometric realization; = Theorem 1.6). *Let S be a finite divisor-closed set of deformation monomials containing 1, and choose a finite labeled set I with at least $n_{a,b}(S)$ labels of each twist occurring in S , with $n_{a,b}(S)$ as in Theorem 1.6. The quotient diagram has finitely many walls, with rays $\rho_1, \dots, \rho_n$ listed in increasing slope. There is a canonical homotopy $(\mathfrak{s}_t)_{t \in [0,1]}$, obtained by concatenating families of simple canonical homotopies in the sense of [GKT22, §3.4], from a family of canonical multisections realizing $\nu_I^{\min}$ to one realizing $\nu_I^{\max}$, the restrictions of the universal extreme indices, such that after reduction to the quotient supported on S every nonzero jump is supported on a single boundary ray, the visible rays occur in increasing slope as t increases, and the total visible jump on ρ_i is the wall automorphism θ_{ρ_i} of Theorem 1.3 modulo that quotient. The choices can be made compatibly over a cofinal construction-closed exhaustion of all labeled graph/descendent data, giving the coefficientwise homotopy stated in Theorem 1.6.*

Proof. Step 1: the labeled model, and symmetrization. Let $\mathfrak{q}_S$ be the monomial ideal spanned by monomials outside S ; it is an ideal because the complement of a divisor-closed set is upward closed. Write a bar for reduction modulo $\mathfrak{q}_S$. Let $A_I = \mathbb{Q}[u_{i,j} : i \in I, j \geq 0] / (u_{i,j}u_{i,j'})$ be the labeled coefficient ring of [GKT22, Def. 4.6], square-zero in each label; write $u_{A,\mathbf{d}} = \prod_{i \in A} u_{i,d_i}$ and $u_i = u_{i,0}$. The monomials $u_{J,\mathbf{d}}$ (indexed by a twist multiset, in the deformation parameters) and $u_{A,\mathbf{d}}$ (indexed by a label subset, in A_I) live in different rings. Symmetrization sends, for every descendent level,

$$t_{a,b,d} \longmapsto \sum_{\substack{i \in I \\ \text{tw}(i) = (a,b)}} u_{i,d}.$$

Let $\mathfrak{q}_{I,S}$ be the labeled monomial ideal whose monomials have symmetric type outside S . Lemma 4.14 of [GKT22] says that symmetrization induces an injection

$$\mathbb{Q}[t_{a,b,d}] / \mathfrak{q}_S \hookrightarrow A_I / \mathfrak{q}_{I,S}.$$

Indeed, a monomial in S maps to the nonempty sum over injective assignments of its occurrences to labels of the required twists; the saturation bound $n_{a,b}(S)$ makes the sum nonempty, and distinct symmetric monomials have disjoint labeled supports. The induced map on the enlarged Lie algebra preserves shifted boundary rays, including the marginal $(1,1)$ -ray. The action after reduction is still free: the minimal-support argument of Lemma 2.4 detects every nonzero class supported in S on W_0 . Thus the $\mathfrak{q}_{I,S}$ -reductions of $\nu_I^{\min}, \nu_I^{\max}$ and of the labeled diagram identify with their universal symmetric counterparts. Below we suppress the labeled-lift symbol in the action of a symmetric factor.

Step 2: the ray factors are symmetric. Twist-preserving permutations fix the two symmetric extreme indices, hence by freeness fix $g^{\min \rightarrow \max}$; they also preserve every boundary-ray subspace. Uniqueness of the factorization in Theorem 1.2(ii), whose filtered proof applies verbatim over the labeled coefficient ring, therefore makes each θ_{ρ_i} symmetric. Write the boundary-ray factorization, in chronological convention, as

$$g^{\min \rightarrow \max} = \theta_{\rho_n} \circ \dots \circ \theta_{\rho_1} \pmod{\mathfrak{q}_{I,S}}, \quad \text{slope}(\rho_1) < \dots < \text{slope}(\rho_n).$$

Step 3: one stage per ray. For each quotient factor choose the symmetric full-group representative $\widehat{\theta}_{\rho_i} = \exp(\widehat{\eta}_{\rho_i})$ whose logarithm is the finite sum of its terms supported in S and has no terms outside S . Starting with a transverse symmetric canonical family having index $\nu_I^{\min}$, construct one stage for each factor. If the current index is ν , then $\widehat{\theta}_{\rho_i}\nu$ is again a symmetric element

of $\Omega_{r,s}(I)$. The realizability statement $\Omega_{r,s}(I) = \Upsilon_{r,s}(I)$ ([GKT22, Cor. 5.13], together with [GKT22, Prop. 3.45]), read in the enlarged sense of Lemma 2.4 on the marginal ray, supplies a symmetric canonical endpoint family with this index. The constructive realization in [GKT22, Thm. 5.12, Lem. 5.14] may be chosen transverse, and twist symmetrization is performed branchwise, preserving transversality. Join the two transverse endpoint families by a family of simple canonical homotopies as in [GKT22, Lem. 3.39].

Step 4: ray purity of each stage. We claim that every jump internal to this stage is supported on ρ_i . For the actual chronological order of its subset times, [GKT22, Thm. 5.8] writes the total labeled jump as

$$\prod_{\emptyset \neq A \subseteq I}^{\text{time}} \exp(v_A),$$

where v_A has exact label support A (it is the sum over descendent profiles of terms $u_{A,\mathbf{d}}X$). In the labeled coefficient ring, $u_{A,\mathbf{d}}u_{B,\mathbf{e}} = 0$ when $A \cap B \neq \emptyset$. For any fixed order this exact-support factorization is unique. Indeed, a nonzero bracket of disjoint supports has support $A \cup B$; hence

$$\log \left(\prod_A \exp(v_A) \right) \Big|_A = v_A + P_A(v_B : B \subsetneq A),$$

and induction on $|A|$ recovers every v_A .

The total jump and the labeled lift of $\widehat{\theta}_{\rho_i}$ carry the same initial endpoint potential to the same final one. Freeness of the enlarged labeled action (Lemma 2.4) makes them equal. Now induct again on $|A|$. The exact- A component of the logarithm of the target is supported on ρ_i ; by induction all v_B for $B \subsetneq A$ are supported there as well. Every Lie word in P_A therefore vanishes because the boundary-ray subalgebra is abelian, as proved in Theorem 1.2(ii). Thus v_A equals the target's exact- A component and is supported on ρ_i . This proves the claim, including for the enlarged torus factor.

Step 5: the correction stage. After the visible stages the resulting full index ν^* is congruent to $\nu_I^{\max}$ in the stated quotient. Let g_* be the unique full transition from ν^* to $\nu_I^{\max}$. Quotient freeness gives $\bar{g}_* = \text{id}$. Join the current family to a transverse symmetric family realizing $\nu_I^{\max}$ by one further Lemma-3.39 stage, chosen as above. Its total jump equals g_* by full freeness. Applying the exact-support induction above *after reduction* to its time-ordered factorization shows successively that every one of its internal jump factors has trivial class: this correction stage is entirely invisible modulo $\mathfrak{q}_{I,S}$.

Step 6: concatenation. Give all stages constant collars at their endpoints, reparametrize them to successive subintervals of $[0, 1]$, and concatenate. Canonicity, strong positivity and simplicity ([GKT22, Defs. 3.17, 3.21, 3.35–3.36]) survive this operation: use identical collars and time reparametrizations simultaneously on every graph and component, which preserves the boundary-factorization identity [GKT22, (3.12)] and the component-independence defining simplicity; transversality is unchanged away from the collars and follows on them from spatial transversality of the endpoint sections, while strong positivity is slicewise. The resulting homotopy has the exact endpoints and the asserted visible order and total jumps. It is not one global Lemma-3.39 normal-form family: the same marking subset may jump in different ray stages. That distinction is precisely what evades Proposition 7.23.

Step 7: compatible choices. The correction just used completes a full finite-label family outside S and therefore should not be nested: a term invisible for S can become visible in a larger lower set. Instead exhaust the countable labeled graph/descendent category by finite *construction-closed* collections $\mathcal{C}_1 \subset \mathcal{C}_2 \subset \dots$. Such a collection contains, with each datum,

all internal-marking subsets, its whole balanced diagonal and critical predecessors, and every component of BA required by a canonical boundary stratum. Closing finitely many data under these operations is finite.

Two relative forms of the GKT constructions apply. First, a transverse canonical endpoint family on $\mathcal{C}_n$ extends to $\mathcal{C}_{n+1}$, fixed on $\mathcal{C}_n$, whenever the target chamber index extends the old one. To see this, run the proof of [GKT22, Thm. 5.12] in its order $(|J|, n(\Gamma), -k_{12}(\Gamma))$ while freezing $\mathcal{C}_n$. Lemma 5.14 of [GKT22] extends each already prescribed boundary datum, and the modifications in (5.51)–(5.53) have exact marking set J ; the observation following (5.52) says that they do not alter earlier marking sets and that distinct descendent vectors have independent top-boundary inductions. Construction-closedness therefore prevents any modification of a frozen multisection. Second, a simple canonical homotopy on $\mathcal{C}_n$ extends relatively. The proof of [GKT22, Lem. 3.39] inducts on $\dim \mathcal{PM}_\Gamma$, prescribes the homotopy on endpoints and canonical boundary strata by [GKT22, (3.14)], and perturbs only the remaining interior. It consequently leaves every frozen homotopy unchanged.

Choose a cofinal sequence of lower sets S_n detected by the $\mathcal{C}_n$. Fix once and for all pairwise disjoint closed intervals $J_\rho \subset (0, 1)$ ordered by the countable set of wall slopes. They exist by inserting the m th interval strictly inside the gap left by the finitely many earlier intervals. At level n , define only the partial endpoint families on $\mathcal{C}_n$, realize its finitely many ray factors on the corresponding J_ρ , and leave every other interval constant. There is no correction: unseen graph data remain undefined. Apply the two relative extensions at level $n + 1$; a new ray is constant on $\mathcal{C}_n$, and a new term on an old ray extends the old stage because the ray factors are compatible under restriction.

Organize the exhaustion simultaneously by finite relabeling isomorphism classes. When a class first appears, choose the new endpoint data on one representative, symmetrize it under its finite twist-preserving stabilizer, and transport it to every later representative; the relative extensions leave all previously transported data fixed. This prescription is imposed only on the endpoints. The intermediate simple homotopies need not be symmetric because their distinct subset times generally break that symmetry.

In the union, every graph/descendent datum is eventually fixed literally. Every such datum depends on only finitely many divisor monomials and boundary degenerations, hence sees only finitely many nonconstant intervals. Constant collars therefore give a genuine smooth canonical homotopy on each $[0, 1] \times \mathcal{PM}_\Gamma$. All canonical boundary identities hold because each finite collection of them occurs at a finite stage. The two endpoints are natural under every finite twist-preserving relabeling and hence pro-symmetric. This is the asserted compatible coefficientwise homotopy; it does not turn the dense descendent ray set into a locally finite chamber decomposition. $\square$

A single unstaged homotopy already suffices below the first degree at which one diagonal carries two walls.

Proposition 7.21 (Exact first two-wall order; unstaged realization). *For $m \geq 1$, the maximum of the numerical diagonal count $N(J, \mathbf{0})$ over primary multisets with $|J| = m$ is*

$$N_{\max}(r, s; m) = m + 1 - \left\lceil \frac{2m}{r} \right\rceil - \left\lceil \frac{2m}{s} \right\rceil,$$

attained by $J = \{(r - 2, s - 2)^m\}$. Consequently the least primary degree admitting a diagonal with two walls is

$$N_2(r, s) = \min \left\{ m \geq 3 : \left\lceil \frac{2m}{r} \right\rceil + \left\lceil \frac{2m}{s} \right\rceil \leq m - 1 \right\},$$

where the minimum of the empty set is ∞ .

- (i) $N_2 = \infty$ if and only if $\hat{c}_W \leq 1$, that is, if and only if W is simple or parabolic; when $\hat{c}_W > 1$, $N_2 \geq \lceil 1/(\hat{c}_W - 1) \rceil$.
- (ii) When $N_2 < \infty$, one unstaged Lemma-3.39 normal-form homotopy realizes the primary diagram modulo $\mathfrak{m}^{N_2}$. When $N_2 = \infty$, such a homotopy exists at every finite order, realizing the completed primary diagram as the inverse system of these finite-order statements; no compatibility of the homotopies themselves is asserted.

Proof. Write $\sum_j a_j = r(J) + r\ell_1$ and $\sum_j b_j = s(J) + s\ell_2$. Since $a_j \leq r - 2$,

$$\ell_1 \leq \left\lfloor \frac{m(r-2)}{r} \right\rfloor = m - \left\lceil \frac{2m}{r} \right\rceil,$$

and similarly for ℓ_2 ; both bounds are attained simultaneously by the maximal twist. Substitution in $N = \ell_1 + \ell_2 - m + 1$ proves the formula for $N_{\max}$ and hence the displayed formula for N_2 . In particular $m = 1, 2$ never gives $N \geq 2$.

The coarser bound from the proof of Theorem 1.1 is $\ell_1 + \ell_2 \leq m\hat{c}_W$, hence

$$(14) \quad N(J, \mathbf{0}) \leq 1 + m(\hat{c}_W - 1),$$

which is at most 1 for every m when $\hat{c}_W \leq 1$, and, when $\hat{c}_W > 1$, forces $m \geq 1/(\hat{c}_W - 1)$ whenever $N \geq 2$; this is (i) apart from the converse. Conversely, if $\hat{c}_W > 1$, the exact maximal-twist count satisfies $N_{\max} > m(\hat{c}_W - 1) - 1$, which is at least 2 for all sufficiently large m , so $N_2 < \infty$.

For (ii), put $K = N_2$ when $N_2 < \infty$ and otherwise fix an arbitrary finite K . Work modulo $\mathfrak{q}_K = (\mathfrak{m}^K, \text{descendent parameters})$ and choose at least $K - 1$ labels of every primary twist, so the symmetrization map used in Theorem 7.20 is injective in the visible degrees. Then every visible diagonal has $N \leq 1$, and a marking subset $A \subseteq I$ determines the single primary diagonal $(A, \mathbf{0})$. Choose transverse symmetric families realizing $\nu_I^{\min}$ and $\nu_I^{\max}$ and take the one Lemma-3.39 normal-form family between them. Its jumps occur at the subset times t_A . By the element displayed in the proof of [GKT22, Thm. 5.8], the reduction of the jump at t_A is a sum over the critical graphs with marking set A : it is zero if $|A| \geq K$ or no visible wall exists, and otherwise is supported on the unique wall of that diagonal.

Order the times of all subsets whose visible primary diagonal carries its unique wall by that wall's slope; place the no-wall and invisible subset times arbitrarily. A resulting zero jump is harmless. This prescription is legitimate: [GKT22, Lem. 3.39] constrains the times only to be distinct in $(0, 1)$, and its construction takes them as given: the zeros are moved onto a prescribed $t_{I(\Gamma)}$ by a diffeomorphism preserving the moduli factor, and the finitely many non-transverse times are kept off the t_A by re-extending transversally at each prescribed time. Visible jumps sharing a ray commute by the abelianness proved in Theorem 1.2(ii). Full freeness identifies the total product with $g^{\min \rightarrow \max}$; after reduction, its grouping by ray is therefore the unique boundary-slope factorization $\prod_{\rho}^{\rightarrow} \theta_{\rho}$. Symmetry and injectivity descend this labeled statement to the universal primary quotient. Thus the total visible jump on ρ_i is θ_{ρ_i} . $\square$

Remark 7.22. The ordering step above is the only place the freedom in the times of [GKT22, Lem. 3.39] is used, and the freedom is real rather than assumed: the lemma imposes on the family $(t_A)_{A \subseteq I}$ only that its members be distinct points of $(0, 1)$, its proof never constructs a t_A but consumes each as a given target, and the induction there runs on $\dim \mathcal{PM}$, which is unrelated to the marking set indexing the times. What the ordering does *not* do is separate the walls of a single diagonal: those all carry the one marking set J , so they share the one time t_J whatever order is chosen, which is Proposition 7.23.

At the first two-wall degree, one global Lemma-3.39 normal-form family provably does not suffice. This explains why the stagewise construction in Theorem 7.20 is necessary.

Proposition 7.23 (One normal-form family cannot separate a diagonal's walls). *Let $(J, \mathbf{0})$ be a primary diagonal with $N \geq 2$ and let H^* be a family of simple canonical homotopies with times $(t_A)_{A \subseteq I}$ as in [GKT22, Lem. 3.39]. Then every possible contribution from the diagonal's N walls is concentrated at the single time t_J , with wall p actually contributing if and only if $\Delta(\Lambda_{J,p}) \neq 0$. The jump there is supported on one ray if and only if at most one of these zero counts is nonzero. In particular no single such normal-form family gives the ray-by-ray realization of Theorem 7.20 once two of the diagonal's walls are actually crossed.*

Proof. The N critical graphs $\Lambda_{J,1}, \dots, \Lambda_{J,N}$ all carry the internal marking set J , so by [GKT22, Lem. 3.39(3)(a)] their zeros occur at the one time t_J . By the jump formula in the proof of [GKT22, Thm. 5.8], the logarithm of the jump there is

$$\sum_{p=1}^N (-1)^{|J|+k_2(\Lambda_{J,p})-1} \Delta(\Lambda_{J,p}) u_{J,\mathbf{0}} X_{k_1(\Lambda_{J,p})-1, k_2(\Lambda_{J,p})-1}.$$

Its boundary directions are pairwise distinct: along p the first boundary degree increases by r and the second decreases by s , both by at least 2 from values at least 1, so the slope strictly decreases; and for $N \geq 2$ none of them is the torus direction. The displayed generators lie in distinct bidegree summands, so no cancellation is possible. Its logarithm is therefore supported on one ray exactly when at most one coefficient is nonzero. $\square$

The stagewise theorem lies outside this family: it applies Lemma 3.39 separately to each boundary-ray factor and may therefore reuse the same marking subset at several stage times. Proposition 7.21 succeeds without staging because when $N \leq 1$ the at-most-one condition on $\Delta(\Lambda_{J,p})$ holds for free.

Example 7.24 (The first two-ray diagonal of $x^5 + y^5$). For $x^5 + y^5$ the order $N_2 = 5$ of Proposition 7.21 is attained by $J = \{(3,3)^5\}$, $\mathbf{d} = \mathbf{0}$: here $m(J, \mathbf{0}) = 50$, the Euler weight is 25, and $N = 2$, with balanced profiles $(0,10), (5,5), (10,0)$ and critical graphs of boundary degrees $(1,6)$ and $(6,1)$: two walls on two *distinct* rays, the first such diagonal for this Fermat pair. Their jump vectors are $w_1 = 30e_1 - 5e_0$ and $w_2 = 5e_2 - 30e_1$. Since $d(J, \mathbf{0}) = -3 < 0$ the chamber-index condition is $\mathcal{A}(J, \mathbf{0}, \nu) = 0$, which reads

$$\frac{6}{25} \nu_{(0,10)} + \frac{1}{25} \nu_{(5,5)} + \frac{6}{25} \nu_{(10,0)} = \frac{204}{15625} :$$

three coefficients against two torsor directions and one condition, as Theorem 1.4 requires. In the minimal normal form of Theorem 1.3 this gives

$$\langle (\tau_0^{(3,3)})^5 \sigma_2^{10} \sigma_{12} \rangle^{\min} = \frac{34}{625},$$

with $\langle (\tau_0^{(3,3)})^5 \sigma_1^{10} \sigma_{12} \rangle^{\max}$ the same by symmetry. (Its agreement with a closed value of Example 6.1 is a numerical accident.) This is the lowest primary degree for $x^5 + y^5$ at which a single unstaged jump would have to carry two rays, hence the smallest test distinguishing Theorem 7.20 from one global Lemma-3.39 normal-form family; every sector computed in §5 and in §7 below has $N \leq 1$, which is why the difficulty does not surface there.

The hypothesis of Proposition 7.23 is met here. Take I to carry exactly five markings of twist $(3,3)$, so that this monomial is realized by a single marking set and the walls above it share one time. Writing $\nu^{\max} - \nu^{\min} = c_1 w_1 + c_2 w_2$ and reading the extreme rows gives $c_1 = c_2 = \frac{34}{3125}$, the middle row being consistent; more generally $c_1 = \nu_0/b_1$ and $c_{p+1} = a_p c_p/b_{p+1}$, so every c_p is nonzero as soon as ν_0 is, exactly as in the proof of Proposition 7.19. No lower-order term can

absorb either coefficient: the walls of this tower sit at $|J| = 2, 4, 5$ and no multiset drawn from the lower orders $|J| = 2, 4$ sums to 5, while a wall acting on a lower coefficient of the potential contributes nothing here, its direction being parallel to that coefficient's and hence annihilating it (Proposition 7.2). So both zero counts are nonzero and both walls are crossed at the one time.

The same computation shows the difficulty is not peculiar to the hyperbolic regime. A single insertion of maximal twist with two descendents has $N = 2$ for every (r, s) , with both coefficients again nonzero, so the obstruction appears in every model once descendents are admitted, the simple ones included, which carry no primary wall at all. What is special about $x^5 + y^5$ is that it is the first symmetric hyperbolic Fermat pair and this is its first primary occurrence.

Corollary 7.25. *For $x^5 + y^5$ modulo $\mathfrak{m}^6$, no family of simple canonical homotopies in one global normal form supplied by [GKT22, Lem. 3.39] between families realizing the extreme indices $\nu^{\min}$ and $\nu^{\max}$ has all of its jumps supported on single rays.*

Proof. By Example 7.24 both walls of the diagonal $J = \{(3, 3)^5\}$ are crossed, and by Proposition 7.23 they share the single time t_J , so the jump there meets two distinct rays. $\square$

Remark 7.26. For simple W the *primary* statement carries no walls at all: $N \leq 0$ throughout by Theorem 1.1, so $\mathfrak{D}_0^{r,s}$ is empty and the direction sector is a single chamber. By Proposition 7.19, though, the descendent walls of the same model are dense in that same sector. The parabolic pairs are the first case with primary content, and there the unique ray is $(1, 1)$, carrying the marginal torus direction; the jump it effects is the one exhibited in [GKT26, Ex. 4.9]. In the hyperbolic regime diagonals with $N \geq 2$ occur and one unstaged jump can then carry several rays at once. The exact first degree is given by Proposition 7.21. Up to the symmetry $(r, s) \leftrightarrow (s, r)$, its complete hyperbolic classification is

r	s	$N_2(r, s)$
3	7	21
3	8	12
3	$9 \leq s \leq 11$	9
3	$s \geq 12$	6
4	5	10
4	$6 \leq s \leq 7$	6
4	$s \geq 8$	4
5	$5 \leq s \leq 7$	5
5	$s \geq 8$	4
$r \geq 6$	$s \geq r$	3

Thus $N_2 \leq 21$ for hyperbolic integral Fermat exponents, with the maximum attained only at $(3, 7)$ up to symmetry. The coarser lower bound in Proposition 7.21 is useful for a fixed pair but does not produce an unbounded sequence approaching the threshold: the exponents are discrete.

Remark 7.27 (Why staging is necessary). On every finite divisor-closed coefficient set, Theorem 7.20 realizes the successive slope cuts by intervals in $[0, 1]$, the two sides of the direction sector by the two endpoints, and the extreme chamber indices by $\nu^{\min}$ and $\nu^{\max}$. In a finite primary quotient these cuts bound actual open chambers; in the descendent completion the

compatible realization is only coefficientwise. The marginal torus factor of Theorem 1.2(ii) is carried by the $(1, 1)$ -ray, whose fixed monomials are exactly its own.

Proposition 7.23 does not contradict the theorem: it concerns one global Lemma-3.39 normal-form family, in which a marking subset has only one jump time. The proof of the theorem instead connects the endpoints of each boundary-ray factor in a separate stage. Exact labeled-support uniqueness forces every internal jump of that stage onto the prescribed ray, and concatenation may reuse a marking subset in later stages. The ray-purity step does not use the single-critical-graph modification of [GKT22, Lem. 5.16] to prescribe a stage jump, nor does it identify that lemma's relative Euler number with the zero count in [GKT22, Thm. 5.8]. (The established endpoint-realizability theorem [GKT22, Cor. 5.13] remains part of the input.)

7.4. Polar extremes and the projective position-space obstruction. The open FJRW invariants of $x^r + y^s$ are coefficientwise weighted broken-line transports on the direction sector, with the closed extended invariants of §6 as vertex data. Theorem 7.11 gives the seed constants an enumerative zero-cycle class. We next give the extreme indices zero-free representatives. This is an existence statement, not a uniqueness statement. We then determine exactly how much position-space geometry the intrinsic boundary-charge sector of Definition 7.1 supports and prove the obstruction to the stronger interpretation.

Call a canonical family σ_2 -*polar* if, on every wall-bearing diagonal, its multisection on each non-extreme balanced graph $\Gamma_{J,p}$, $p \geq 1$, is zero-free; call it σ_1 -*polar* if the same holds for $p \leq N(J, \mathbf{d}) - 1$.

Lemma 7.28 (Relative nonvanishing). *Let X be a connected compact n -dimensional orbifold with corners, $n \geq 2$, and let $E \rightarrow X$ be a rank- n real orbibundle relatively oriented with respect to X . Let A be a sufficiently small collar of the full boundary (and take $A = \emptyset$ when the boundary is empty), and let s_A be a smooth nowhere-zero multisection on A . Then s_A extends, fixed on a smaller collar, to a nowhere-zero smooth multisection of E if and only if*

$$\deg e(E; s_A) = 0.$$

The same conclusion holds on a connected compact core of a GKT partial compactification, with A the collar of its nonexchange and cutoff boundary.

Proof. Choose an orbifold metric, delete the zero section, and normalize radially. The extension problem is then one for the sphere bundle $S(E)$, with fiber S^{n-1} . Take a finite relative orbifold triangulation of (X, A) subordinate to the corner strata and bundle charts. Since S^{n-1} is $(n-2)$ -connected, branches extend through the relative $(n-1)$ -skeleton. On a cell with finite stabilizer, replace a branch by its equally weighted stabilizer orbit; on overlaps take a common branch refinement. These operations preserve nonvanishing.

The only obstruction is consequently the top obstruction

$$o(s_A) \in H^n(X, A; \mathfrak{o}_E \otimes \mathbb{Q}),$$

which the usual top-cell clutching calculation identifies with the relative Euler class. Poincaré–Lefschetz duality, relative orientation, and connectedness give

$$H^n(X, A; \mathfrak{o}_E \otimes \mathbb{Q}) \cong H_0(X; (\mathfrak{o}_E \otimes \mathfrak{o}_X) \otimes \mathbb{Q}) \cong \mathbb{Q}.$$

Evaluation on the relative fundamental class is therefore an isomorphism, so the displayed degree vanishes exactly when the obstruction does. Extending over the top cells and smoothing after a common equivariant subdivision gives the required multisection. Conversely, a nowhere-zero extension is a sphere-bundle multisection and has zero relative Euler class. $\square$

Theorem 7.29 (Polar representatives of the extremes). *For every finite internal marking set I , the index $\nu_I^{\min}$ admits a transverse symmetric σ_2 -polar canonical representative and $\nu_I^{\max}$ admits a transverse symmetric σ_1 -polar canonical representative. Conversely, the chamber index of every σ_2 -polar family is $\nu^{\min}$, and that of every σ_1 -polar family is $\nu^{\max}$. The finite choices may be made compatibly over a cofinal construction-closed exhaustion.*

Proof. A zero-free multisection has zero weighted count. A σ_2 -polar family therefore has $\nu_{J,p} = 0$ for every $p \geq 1$ on every wall-bearing diagonal. The uniqueness part of Theorem 1.3 identifies it with $\nu^{\min}$. The other assertion is symmetric. Diagonals with $N = 0$ impose no polar condition and are fixed by the chamber-index equations.

It remains to prove existence. We construct the σ_2 -polar family; the other construction exchanges the two boundary types and reverses p . Run the construction of [GKT22, Thm. 5.12] with target $\nu_I^{\min}$, but refine its extension step as follows. Fix a wall-bearing diagonal $(J, \mathbf{d})$ and a balanced graph $\Gamma_{J,p}$ with $p \geq 1$. Then

$$\nu_{J,p}^{\min} = 0, \quad n := \dim \mathcal{PM}_{\Gamma_{J,p}} = k_1 + k_2 + 2|J| - 2 \geq 2;$$

indeed, $J \neq \emptyset$ and $k_1 = r(J) + pr \geq r$.

Temporarily label every boundary marking and fix the relative cyclic order of the r -labels and, separately, of the s -labels. The components in this block are indexed by the binary shuffles of k_1 letters R and k_2 letters S . Two shuffles differing by one adjacent interchange $RS \leftrightarrow SR$ meet exchangeable boundary faces. The graph on shuffles with these edges is connected. Glue each exchange pair. This is the gluing described in [GKT22, Rem. 1.54]; the exchange map reverses the induced boundary orientation by [GKT22, Prop. 2.54]. The result is a connected relatively oriented orbifold with corners $\widehat{\mathcal{PM}}_{J,p}$. Its remaining boundary consists of the nonexchange boundary and the cutoff near the positive ends. If $k_2 = 0$, the shuffle set is a singleton and there are no exchange faces; the same notation denotes the original connected component and the argument below applies directly.

The descendent Witten bundle glues as well. For exchange-equivalent faces Λ and Λ' , [GKT22, Obs. 3.9] gives

$$F_{\Lambda'} \circ \text{Exchange} = F_{\Lambda},$$

so the canonical boundary condition of [GKT22, Def. 3.21] has identical traces under the gluing. Before gluing, the zero-cycle on a fully labeled shuffle block is the signed sum over one lift of every R/S word. Relabeling within either boundary type preserves the moduli and bundle orientations [GKT22, Lemmas 2.44(2), 2.46(2) and Prop. 2.51(3)]. Consequently, for a positive rational covering factor c_{k_1, k_2} ,

$$(15) \quad \deg e(\widehat{E}; \widehat{s}) = c_{k_1, k_2} \left\langle \prod_{i \in J} \tau_{d_i}^{(a_i, b_i)} \sigma_1^{k_1} \sigma_2^{k_2} \sigma_{12} \right\rangle^{s, o} = 0.$$

Gluing changes no interior local degree; Proposition 2.54 supplies precisely the sign needed at the paired faces.

Apply Lemma 7.28 while holding fixed the already assembled strongly positive data on the nonexchange boundary and positive-end collars. It gives a nowhere-zero multisection on each exchange-completed shuffle block. Cutting the block open produces nowhere-zero branches on all cyclic-order components and matching traces on every exchange pair. The three-point exchange bubble is rigid. Its detaching map identifies an exchange face, up to its finite label and ghost symmetries, with the smooth $k_{12} = 2$ base. Take the same weighted finite orbit of the entire collection of labeled block sections and push it to the unlabeled moduli. Its seam restriction is exactly the pullback of the corresponding pushed-down multisection on the base.

This is a union of branches, not their vector average, and hence both multisections remain nowhere zero. Since the construction was relative at the corners, all higher boundary-assembling identities are unchanged.

These graphwise choices form one canonical family. The induction in [GKT22, Thm. 5.12] orders graphs by the internal marking set, then by increasing

$$n(\Gamma) = sk_1(\Gamma) + rk_2(\Gamma) + (r + s)k_{12}(\Gamma),$$

and, at fixed n , by decreasing k_{12} . When exchange faces are present, Equation (5.48) there places the $k_{12} = 2$ exchange base before its $k_{12} = 1$ balanced graph. More generally, each additional stable vertex without internal markings has weighted count at least $r + s$ by the divisibility relations (5.50) there. Thus (5.48) shows that a graph depending on this base has larger n , or has the same n and smaller k_{12} ; it occurs later in the order. Treat the base and the exchange completion as one induction block; Lemma 5.14 of [GKT22] supplies the later extensions. Changing this exchange datum leaves every chamber coordinate unchanged: the two contributions have the same pullback by Observation 3.9 and opposite boundary orientations by Proposition 2.54, exactly as in [GKT22, Lem. 5.3]. Thus the ordinary critical-graph modifications still impose $\nu_I^{\min}$, while the refined extensions make every $p \geq 1$ section zero-free.

Finally take the finite weighted orbit under the twist-preserving permutations of I , as in [GKT22, Obs. 3.43]. Every orbit branch is zero-free on the required graphs, so symmetry preserves polarity. Running the same relative construction over the construction-closed exhaustion used in Theorem 1.6, freezing all previous collars and sections, gives the stated coefficientwise compatibility. $\square$

Remark 7.30 (Why scalar vanishing is sufficient only after exchange completion). For fixed traces on every exchange face, the old componentwise warning remains valid. For example, on the first two-wall diagonal of $x^5 + y^5$ the middle profile $(k_1, k_2) = (5, 5)$ has

$$\#\pi_0(\mathcal{PM}_{\Gamma_{J,1}}) = \binom{10}{5} = 252$$

by [GKT22, Rem. 2.27]; one zero scalar sum would not force 252 fixed-boundary relative Euler degrees to vanish. The theorem does not make that inference. It varies the exchange traces coherently and joins the 252 shuffle components before applying the relative obstruction calculation.

We now turn to the position-space question. The lattice, cone, axes, ray placement, determinant pairing and primitive-form shift are intrinsic boundary-charge data. The lattice even supplies natural affine normalizations, for example total boundary charge. The issue is not the existence of a coordinate called radius, but whether continuation detects it and whether the cone supports the required singular affine monodromy.

Put $C_\partial^\times = C_\partial \setminus \{0\}$. For a finite divisor-closed coefficient set S , write $\mathfrak{D}_S$ for the finite union of wall rays in the quotient. A *tame path* has off-wall endpoints and is piecewise smooth and transverse to $\mathfrak{D}_S$, apart from finitely many generic tangencies; tame homotopies are taken relative endpoints and avoid the deleted vertex. Let $\Pi_{\leq 2}^{\text{tame}}(C_\partial^\times, \mathfrak{D}_S)$ denote the resulting path 2-groupoid. Let Can_S be the continuation bicategory whose objects are transverse canonical multisection families reduced to S , whose one-morphisms are collared concatenations of simple canonical homotopies, and whose two-morphisms are relative two-parameter canonical families modulo relative three-parameter families.

Theorem 7.31 (Projective coherent sector and no-go theorem; = Theorem 1.7). *Let S be finite and divisor-closed.*

- (1) *There is a homotopy-coherent pseudofunctor*

$$\mathcal{F}_S : \Pi_{\leq 2}^{\text{tame}}(C_\partial^\times, \mathfrak{D}_S) \longrightarrow \text{Can}_S$$

which sends a positive elementary crossing of a ray ρ to a geometric stage with total jump $\theta_{\rho,S}$, and a negative crossing to its reverse. It is canonical up to coherent equivalence.

- (2) *If*

$$\pi : C_\partial^\times \longrightarrow \mathbb{P}(C_\partial) \cong [0, 1]$$

is projectivization, then

$$(16) \quad \mathcal{F}_S \simeq \overline{\mathcal{F}}_S \circ \pi_*$$

for a coherent transport pseudofunctor on the interval marked by the wall slopes. Thus radial paths act by the identity and $\mathcal{F}_S \circ (\delta_\lambda)_ \simeq \mathcal{F}_S$ for every positive dilation δ_λ . The factorizations are compatible under restriction and define a coefficientwise pro-functor.*

- (3) *The punctured sector has no origin loop. The total product $g^{\min \rightarrow \max}$ is boundary-to-boundary transport, not holonomy about the vertex. The disk fillers in (i) realize backtracking and composition coherence away from the vertex, but no codimension-two joint comparing two orders of distinct rays.*

Consequently a GKT-determined realization, one using no valuation, period, stability, or degeneration datum beyond the present continuation theory, cannot supply a radially detected modulus or a singular affine origin with monodromy. In particular the marginal torus factor is not thereby identified with integral-affine holonomy. A genuine Gross–Siebert/SYZ position space requires additional input.

Proof. Write $q = x_1 e_1 + x_2 e_2$ and set

$$R(q) = x_1 + x_2, \quad a(q) = \frac{x_2}{x_1 + x_2}.$$

Then

$$(17) \quad (R, a) : C_\partial^\times \xrightarrow{\sim} (0, \infty) \times [0, 1], \quad (r, t) \longmapsto r((1-t)e_1 + te_2).$$

If a_i is the slope coordinate of a wall ρ_i , then $\rho_i \setminus \{0\} = (0, \infty) \times \{a_i\}$. Thus the finite wall stratification is the product of a radial half-line with a marked interval. The homotopy

$$(r, t) \longmapsto ((1-u)r + u, t), \quad 0 \leq u \leq 1,$$

is a stratum-preserving strong deformation retraction to $R = 1$.

The crossing word of a tame path is therefore the crossing word of its angular projection. Under a generic homotopy the word changes only by inserting or deleting $e_i e_i^{-1}$ or $e_i^{-1} e_i$: a tangency to the i th marked slope creates or cancels that pair. Crossings of distinct slopes cannot exchange order. Equivalently, the chamber-adjacency graph is a finite path, hence a tree, and every word reduces uniquely to the monotone edge path between its endpoint chambers.

Choose from Theorem 7.20 the intermediate canonical families $\mathfrak{s}_0, \dots, \mathfrak{s}_n$ and the ray stages $h_i : \mathfrak{s}_{i-1} \rightarrow \mathfrak{s}_i$. Assign $\mathfrak{s}_i$ to the i th chamber, h_i to e_i , h_i with reversed parameter to e_i^{-1} , and a collared concatenation to a crossing word. The loop $h_i * h_i^{-1}$ has compatible canonical data on the boundary of a parameter disk. Lemma 7.12 fills it relative to that boundary. The same lemma fills relations among several cancellations; the multiplihedra used in Theorem 7.14 give all parenthesization coherences. Applying the relative extension once more joins two choices of representatives, stages, collars, or fillers. This proves (i).

The assignment just constructed uses only the angular chamber and the oriented crossing word. It is therefore the pullback of the interval construction, which proves (16); a radial path has constant projection and $\pi \delta_\lambda = \pi$. If $S \subset T$, uniqueness of the ray factorization gives

$\theta_{\rho,T}|_S = \theta_{\rho,S}$, and the construction-closed relative exhaustion in Theorem 7.20 extends the old stages and fillers without changing them. Every fixed coefficient sees only finitely many divisor factors. This proves (ii), including the pro-statement.

Finally, (17) makes $C_\partial^\times$ contractible. Its two boundary sides are not identified, so the slope-ordered path from the e_1 -side to the e_2 -side is not a loop. Distinct walls are disjoint after deletion of the vertex, and the crossing-word argument shows that the available disks cannot compare their two orders. To create an origin loop one must enlarge the sector to a surface with an interior puncture or glue its two sides. Either operation requires affine transition data, including a linear part in $\mathrm{GL}(N_\partial)$, which the formal wall automorphisms do not supply. This proves (iii) and the final assertion. $\square$

The theorem resolves the intrinsic version of the position-space question negatively while retaining its strongest positive part: the punctured sector carries coherent geometric transport, but that transport is projective. In a genuine tropical Landau–Ginzburg model, the input includes a singular integral-affine manifold, a polyhedral decomposition and a multivalued convex piecewise-linear function, and broken lines use that actual affine geometry [CPS24]. None of those radial or vertex data is recovered by (16).

The same boundary applies to theta functions. The present results transport potentials and their seed decomposition, but they do not construct the binary interacting-broken-line or pair-of-pants operation needed to identify theta multiplication constants with the open invariants. Such theta functions (in the sense of [GHK15, GS19]) could still span the Saito–Givental deformation of $x^r + y^s$ and recast the open mirror theorem [GKT22, Thm. 0.3] as intrinsic mirror symmetry. Theorem 7.31 does not rule out that enriched theory; it proves that it requires new input, for example a Novikov/energy valuation, a period normalization, or a logarithmic degeneration with specified affine gluing. The suggested identification of the marginal torus with monodromy of the elliptic family underlying Maher’s modularity [Mah24] is likewise an additional comparison theorem, not a consequence of the present sector.

8. VERIFICATION

All claims are checked by exact polynomial and rational arithmetic (no floating point); the programs, with replication instructions, are available at <https://github.com/bwuebben/open-fjrw-scattering>. `src/gkt_algebra.py`: the algebra of [GKT22, Def. 4.22] (bracket closure, the classification of Definition 2.1 against brute-force enumeration, the first walls of $x^4 + y^4$ and $x^5 + y^5$, the period-invisibility of the wall action; Theorems 1.1 and 1.4(i)). `src/diagonal_wall_count.py`: the exact maximum $N_{\max}$ and the classification of N_2 in Proposition 7.21, checked against direct multiset enumeration. `src/first_two_ray_diagonal.py`, `src/two_ray_forced.py`: the diagonal of Example 7.24, its $\mathcal{A}$ -condition and normal form, and the nonvanishing of both zero counts (Corollary 7.25). `src/ray_density_exact.py`, `src/ray_accumulation.py`: the ray formula and slope density of Proposition 7.19. `src/oscillatory_check.py`: the seed relations (4) from [GKT22, Thm. 0.3] directly (Remark 4.3). `src/a_invariants.py`: the labeled-convention $\mathcal{A}(J, \mathbf{d}, \nu)$, the convention pin, the tower of Remark 4.2. `src/canonical_diagram.py`: normal forms and the wall functions of Theorem 1.3 (Example 5.1), the $t_{3,3}^4$ relation, the pure- τ invariant of (3). `src/scattering.py`: the wall maps [GKT22, Lem. 4.20] and the (1,1)-shift dictionary (Proposition 2.2). `src/rspin.py`, `src/taut_m0n.py`, `src/closed_fjrw.py`: the closed data of Example 6.1 with its internal consistency checks. `src/verify_thm05.py`, `src/verify_thm05_full.sage`: the full numerical verification of [GKT22, Thm. 0.5]. `src/mirror_periods.py`: Lemma 4.1. `src/anchor_induction.py`, `src/backscatter.py`: the randomized verifications of Theorems 7.7 and 7.5, the

middle-seed tower of Example 7.17, and the far-side enumeration. `src/transport_engine.py`, `src/mixed_sector.py`, `src/farside_test.py`, `src/canonical_seeds.py`: the family and far-side verifications of §7 and the canonical-seed mechanism of Theorem 1.5. `src/seed_chain_formula.py`: the determinant weights, exponential factorial, and the 10/3 far-side term in Theorems 7.10–7.11. All Python verifications complete in minutes on commodity hardware; the SageMath verification is the longest run.

REFERENCES

- [BCT24] A. Buryak, E. Clader, R. J. Tessler, *Open r -spin theory II: The analogue of Witten’s conjecture for r -spin disks*, J. Differential Geom. 128 (2024), no. 1, 1–75; arXiv:1809.02536.
- [CPS24] M. Carl, M. Pumperla, B. Siebert, *A tropical view on Landau–Ginzburg models*, Acta Math. Sin. (Engl. Ser.) 40 (2024), no. 1, 329–382; arXiv:2205.07753.
- [GKT22] M. Gross, T. L. Kelly, R. J. Tessler, *Open FJRW theory and mirror symmetry*, arXiv:2203.02435v2 (2022).
- [GKT26] M. Gross, T. L. Kelly, R. J. Tessler, *Open enumerative geometries for Landau–Ginzburg models*, arXiv:2602.12707v1 (2026).
- [GHK15] M. Gross, P. Hacking, S. Keel, *Mirror symmetry for log Calabi–Yau surfaces I*, Publ. Math. Inst. Hautes Études Sci. 122 (2015), 65–168.
- [GPS10] M. Gross, R. Pandharipande, B. Siebert, *The tropical vertex*, Duke Math. J. 153 (2010), 297–362.
- [Gro10] M. Gross, *Mirror symmetry for $\mathbb{P}^2$ and tropical geometry*, Adv. Math. 224 (2010), 169–245.
- [GS19] M. Gross, B. Siebert, *Intrinsic mirror symmetry*, arXiv:1909.07649 (2019).
- [HLSW22] W. He, S. Li, Y. Shen, R. Webb, *Landau–Ginzburg mirror symmetry conjecture*, J. Eur. Math. Soc. 24 (2022), no. 8, 2915–2978; arXiv:1503.01757.
- [KS06] M. Kontsevich, Y. Soibelman, *Affine structures and non-Archimedean analytic spaces*, in *The unity of mathematics*, Progr. Math. 244, Birkhäuser (2006), 321–385.
- [LLSS17] C. Li, S. Li, K. Saito, Y. Shen, *Mirror symmetry for exceptional unimodular singularities*, J. Eur. Math. Soc. 19 (2017), 1189–1229; arXiv:1405.4530.
- [Mah24] R. Maher, *Predictions in open Fan–Jarvis–Ruan–Witten theory via mirror symmetry, modularity, and wall-crossing*, Ph.D. thesis, University of Birmingham, 2024.
- [MW10] S. Ma’u, C. Woodward, *Geometric realizations of the multiplihedron and its complexification*, Compos. Math. 146 (2010), no. 4, 1002–1028; arXiv:0802.2120.
- [MW17] D. McDuff, K. Wehrheim, *Smooth Kuranishi atlases with isotropy*, Geom. Topol. 21 (2017), 2725–2809; arXiv:1508.01556.
- [Wue26] B. J. Wuebben, *The central-charge threshold for wall-crossing in two-variable open FJRW theory*, companion note, <https://github.com/bwuebben/open-fjrw-scattering> (2026).

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